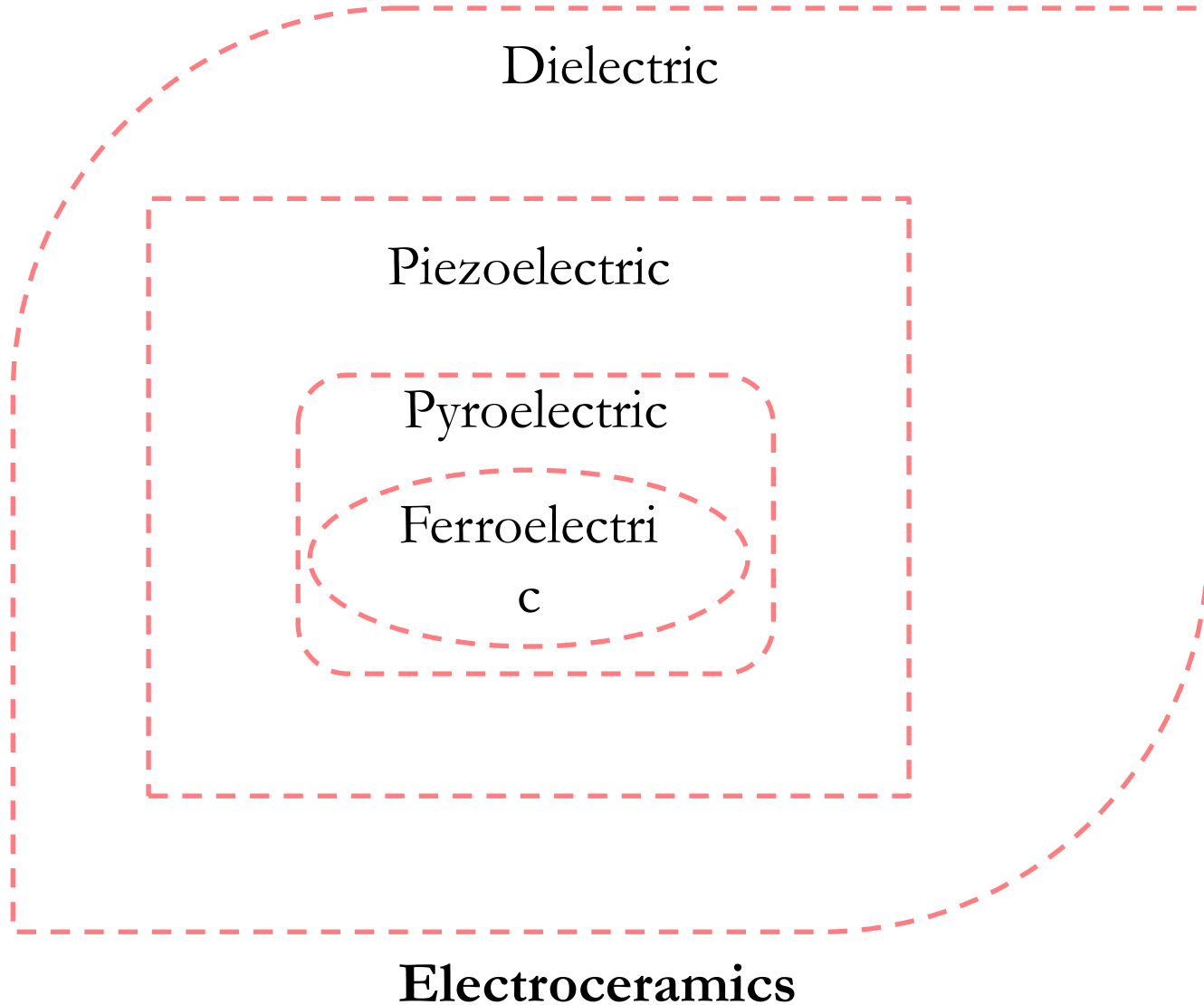


Chapter 11: Electrical Properties

To be covered...



Dielectric Ceramics

📖 A dielectric ceramic is any ceramic material that supports charge without conducting it to a significant degree. In principle all insulators are dielectric, although the capacity to support charge varies greatly between different insulators.

ρ is high, E_g is large



Dielectric materials are used in many applications, from simple electrical insulation to sensors and circuit components.

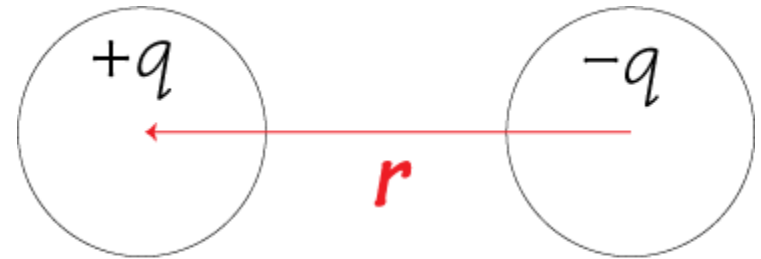
Part I Dielectric Properties

Aims:

- Understand the meaning of the terms dielectric constant, dielectric loss and dielectric breakdown;
- Recognize that the properties of dielectrics are due to polarization and understand how this polarization arises on the microscopic scale;
- Understand how material structure, temperature and frequency affect the properties of dielectrics.
- Be aware of some practical applications of dielectric materials.

Electric dipoles

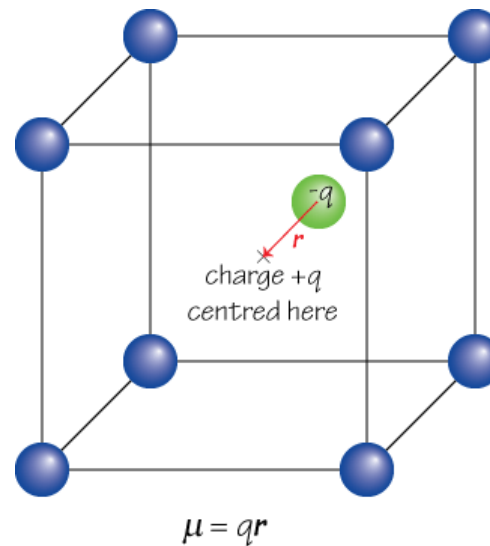
1. If two discrete charged particles of opposite charges are separated by a certain distance, a dipole moment μ arises.



$$\mu = qr$$

2. If the center of positive charge within a given region and the center of negative charge within the same region are not in the same position, a dipole moment μ arises.

For example, in the diagram below the center of positive charge from the 8 cations shown is at X, while the center of negative charge is located some distance away on the anion.



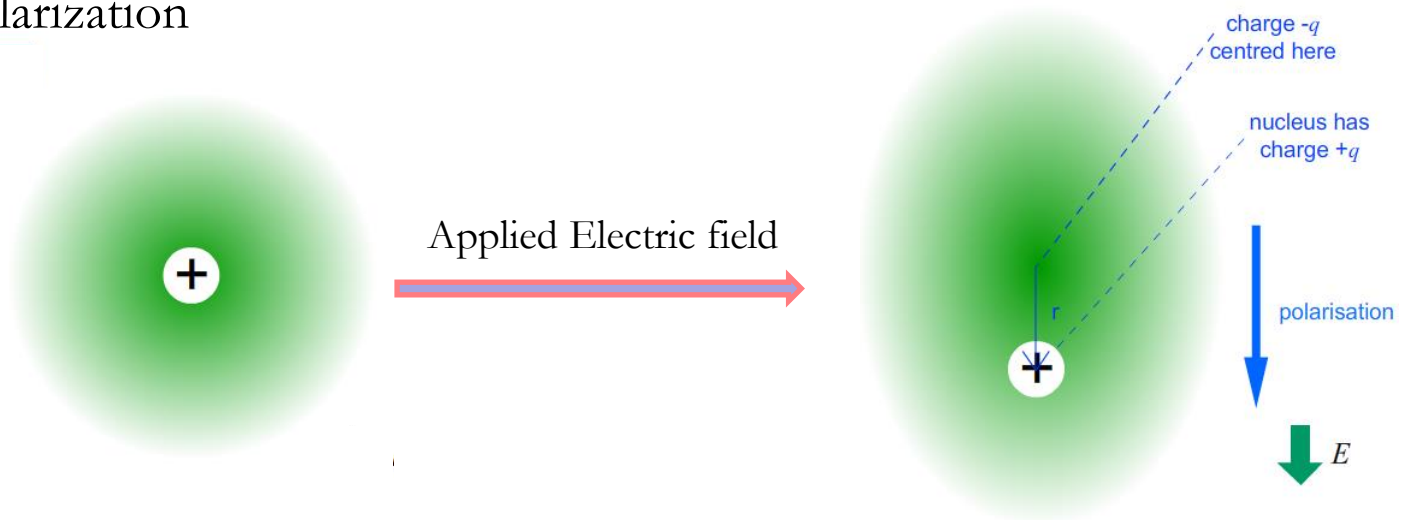
Key

 = cation

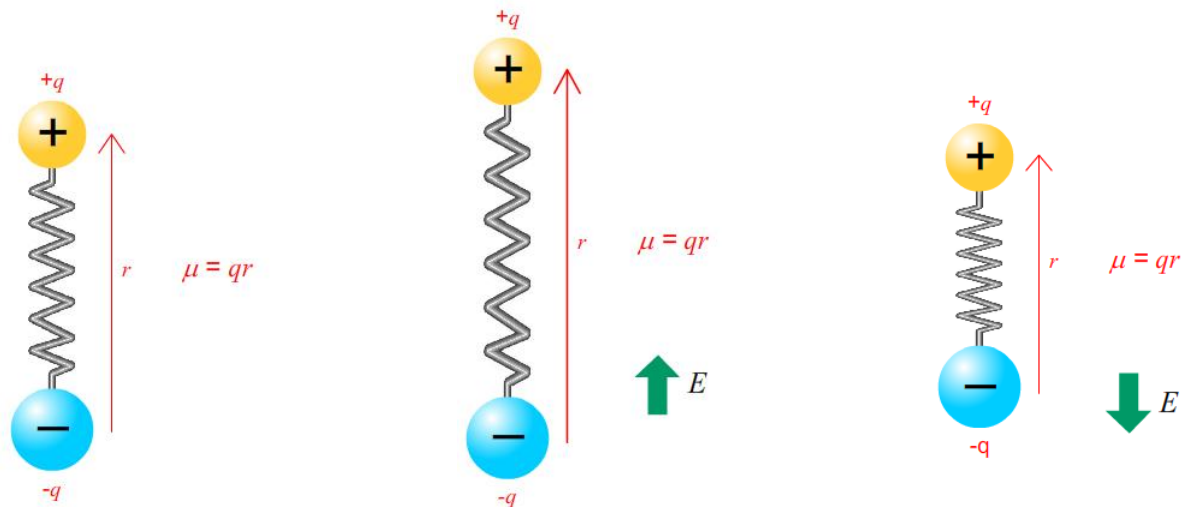
 = anion

Polarization Mechanisms

Electronic polarization

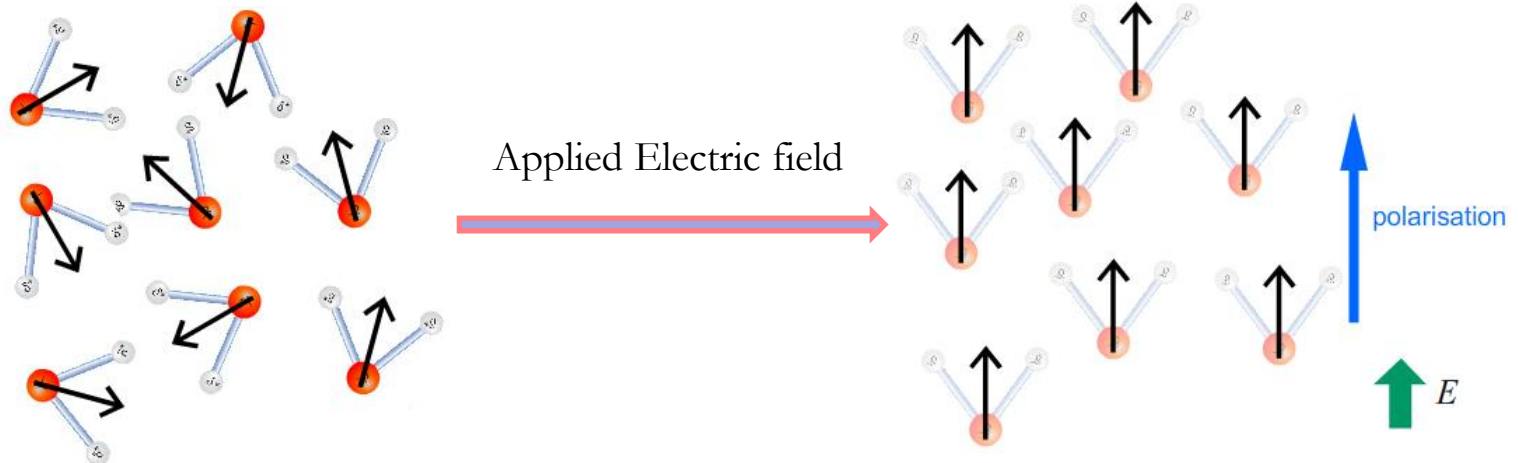


Ionic polarization

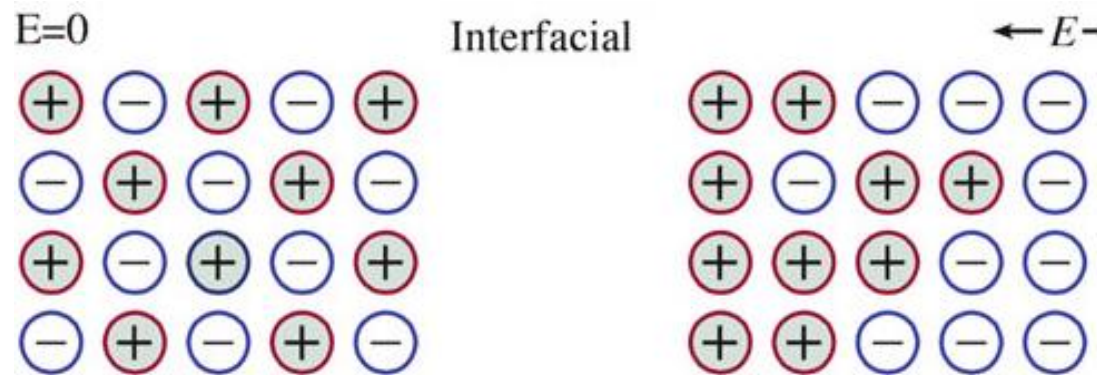


Polarization Mechanisms

Orientational Polarization (Dipolar)

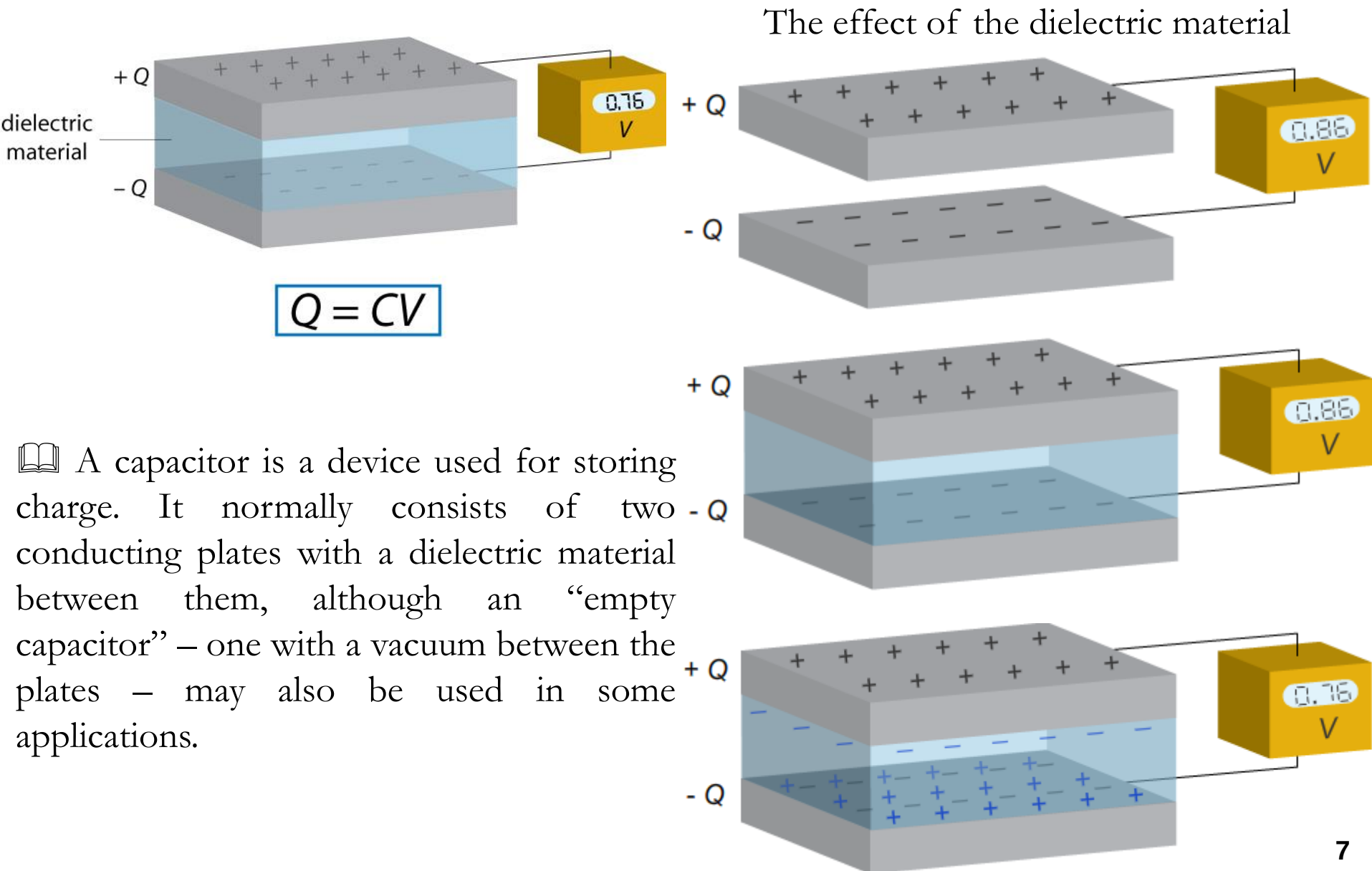


Interfacial Polarization (Space charge)



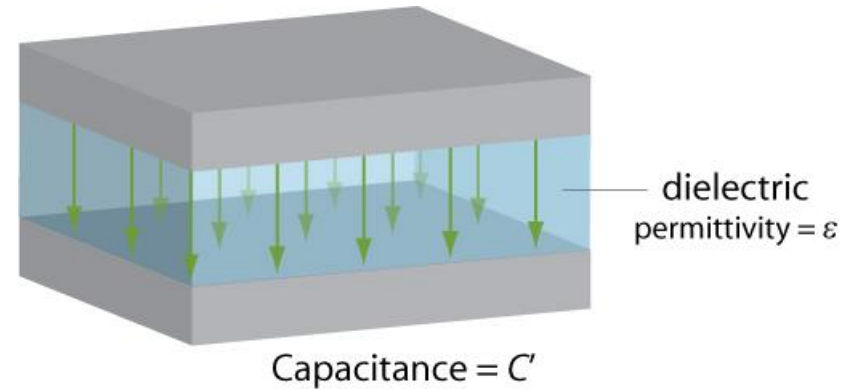
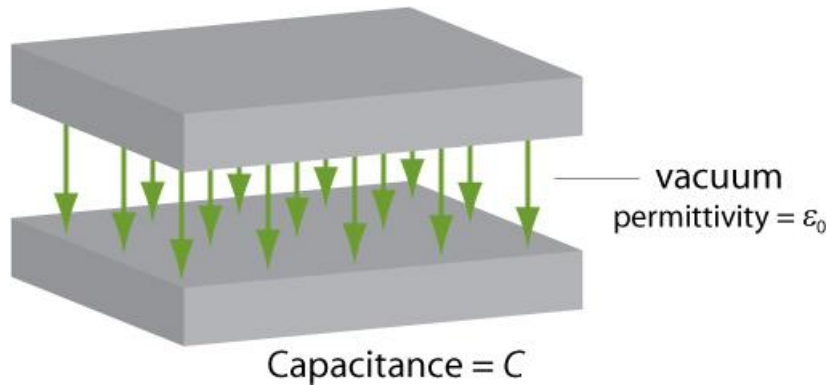
Total P for the material: $P = P_{\text{electronic}} + P_{\text{ionic}} + P_{\text{dipolar}} + P_{\text{interfacial}}$

Capacitors



The dielectric constant

 The **dielectric constant** of a material provides a measure of its effect on a capacitor. It is the **ratio of the capacitance of a capacitor containing the dielectric to that of an identical but empty capacitor.**



$$\kappa = C'/C$$

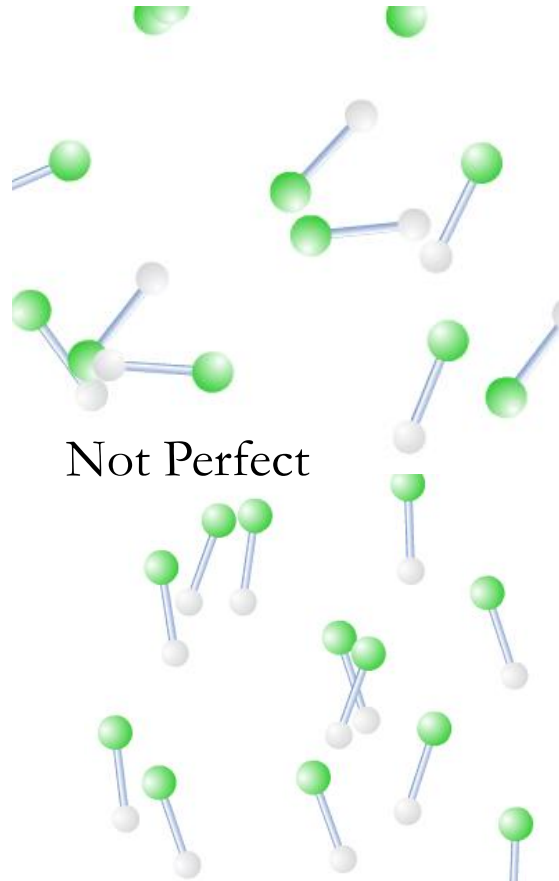
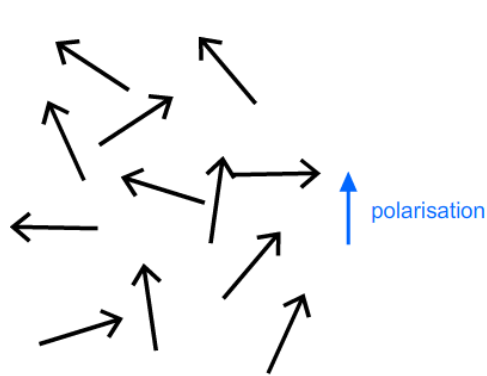
$$\kappa = \epsilon/\epsilon_0$$

 The dielectric constant relates to the permittivity of the material. **Permittivity** (介电常数) is a quantity that describes the effect of a material on an electric field: the higher the permittivity, the more the material tends to reduce any field set up in it. Since the dielectric material reduces the field by becoming polarized, the permittivity expresses the ability of a material to polarize in response to an applied field. **The dielectric is the ratio of the permittivity of the dielectric to the permittivity of a vacuum.**

Effect of temperature on the dielectric constant

For materials that possess permanent dipoles, there is a significant variation of the dielectric constant with temperature, **why?**

Due to the effect of heat on orientational polarization.



As the $T \uparrow$, thermal energy $\uparrow$, amplitude of random thermal motion $\uparrow$, deviation from perfect alignment greater, less closely aligned, dielectric constant $\downarrow$.

Perfect alignment

Not Perfect

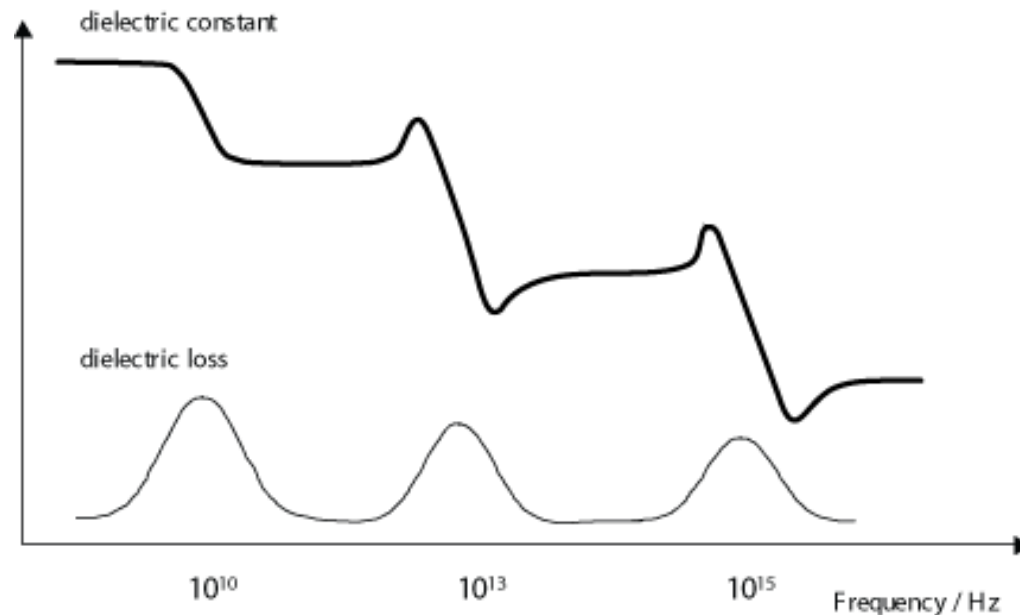
Ideal case

Real case

Loss in dielectrics

An efficient dielectric supports a varying charge with minimal dissipation of energy in the form of heat.

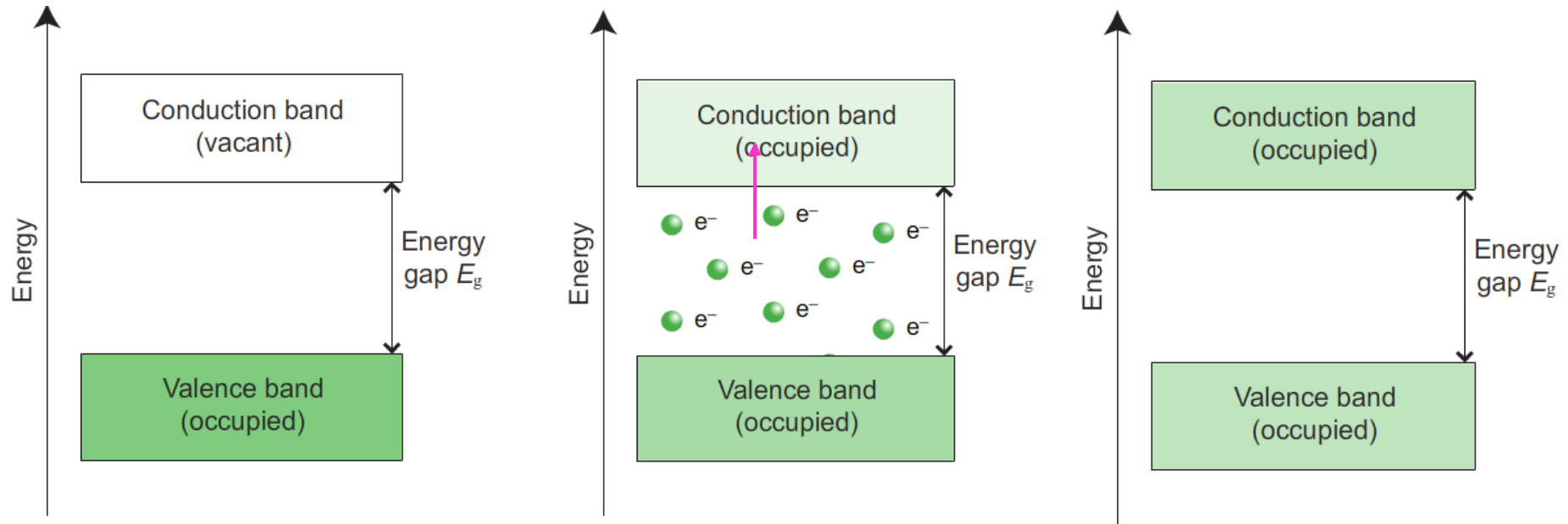
Two main forms of loss : 1) **In conduction loss**, a flow of charge through the material causes energy dissipation; 2) **Dielectric loss** is the dissipation of energy through the movement of charges in an alternating electromagnetic field as polarization switches direction.



Dielectric loss is especially high around the relaxation or resonance frequencies of the polarization mechanisms as the polarization lags behind the applied field, causing an interaction between the field and the dielectric's polarization that results in heating.

Dielectric strength

 **Dielectric breakdown:** At high electric fields, a material that is normally an electrical insulator may begin to conduct electricity – i.e. it ceases to act as a dielectric.



 **The dielectric strength** (breakdown strength): the magnitude of an electric field necessary to produce breakdown.

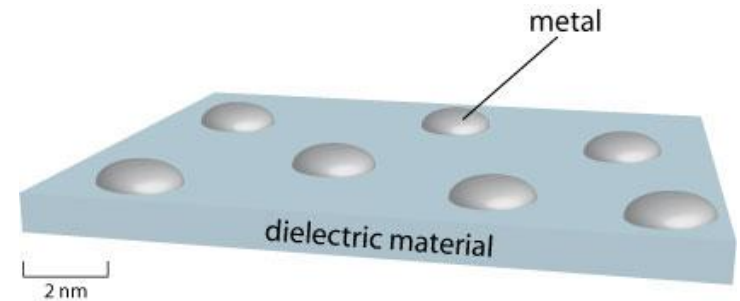
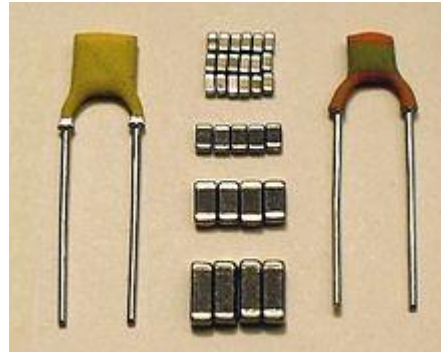
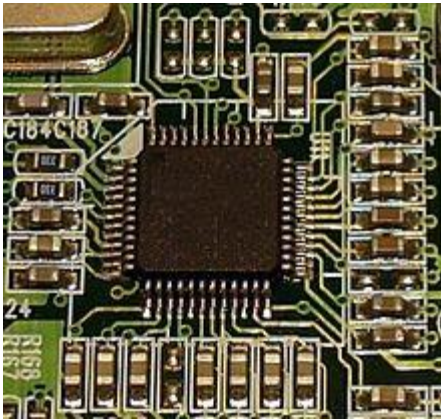
Dielectric constants and dielectric strength of various ceramics

<i>Material</i>	κ at 1 MHz	<i>Material</i>	κ at 1 MHz
Diamond	5.5–6.6	Al ₂ O ₃	8.8
SiO ₂	3.7–3.8	MgO	9.6
NaCl	5.9	BaTiO ₃	3,000
Mica	5.4–8.7	Pyrex glass	4.0–6.0
Soda-lime glass	7.0–7.6	TiO ₂	14–110
Steatite (SiO ₂ + MgO + Al ₂ O ₃)	5.5–7.5	Forsterite (2MgO.SiO ₂)	6.2
Cordierite (SiO ₂ + MgO + Al ₂ O ₃)	4.5–5.4	Mullite	6.6
High-lead glass	19	Vycor glass	3.9

<i>Material</i>	<i>Dielectric Strength (MV/cm at 25° C)</i>
Al ₂ O ₃ (99.5%)	0.18
Al ₂ O ₃ (94.0%)	0.26
High-voltage porcelain	0.15
Steatite porcelain	0.10
Lead glass	0.25
Lime glass	2.5
Borosilicate glass	5.8
Fused quartz	6.6
Quartz crystal	6.0
NaCl [100], [111], [110]	2.5, 2.2, 2.0
Muscovite mica	10.1

Application of dielectric ceramics

A major use of dielectrics is in fabricating **capacitors**. These have many uses including **storage of energy** in the electric field between the plates, filtering out noise from signals as part of a **resonant circuit**, and **insulation wires, cables**, etc.



Strain gauge

Sensor devices: Strain gauge by evaporating a small amount of metal onto the surface of a thin sheet of dielectric material, quantum tunneling.

Summary I

- Dielectrics are electrical insulators that support charge. The properties of dielectrics are due to **polarization**.
- There are **four main mechanisms** : electronic (distortion of the electron cloud in an atom), ionic (movement of ions), orientational (rotation of permanent dipoles) and interfacial polarization.
- A capacitor is a device that stores charge, usually with the aid of a dielectric material. Its capacitance is defined by **$Q = CV$**
- The **dielectric constant** κ indicates the ability of the dielectric to polarize. It can be defined as **the ratio of the dielectric's permittivity to the permittivity of a vacuum**.
- Each of the polarization mechanisms has a characteristic relaxation or **resonance frequency**. In an alternating field, at each of these (materials dependent) frequencies, **the dielectric constant will sharply drop**.
- Polar dielectrics show **a decrease in the dielectric constant as temperature increases**.
- **Dielectric loss** is the absorption of energy by movement of charges in an alternating field, and is particularly high around the relaxation and resonance frequencies of the polarization mechanisms.
- Sufficiently high electric fields can cause a material to undergo **dielectric breakdown** and become conducting.

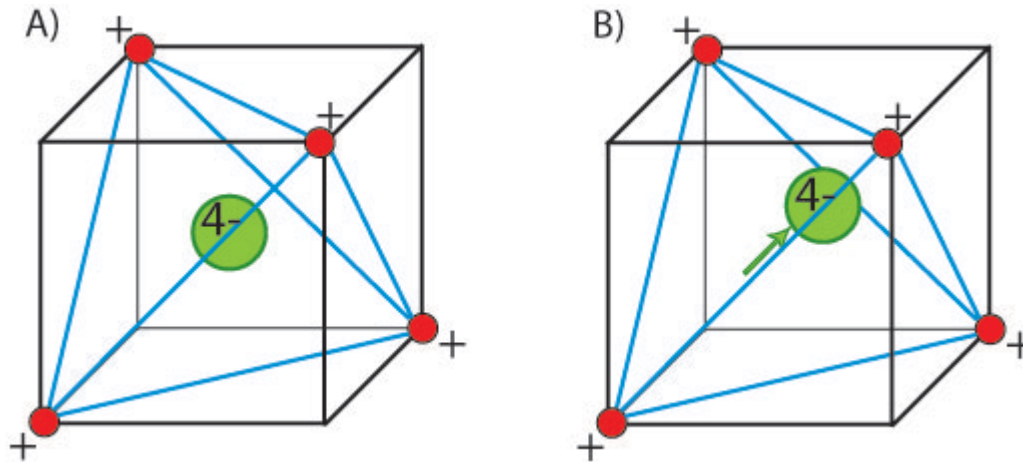
Part II Ferroelectric Properties

Aims:

- Be aware of the atomic requirements for Ferroelectricity.
- Understand how this atomic structure leads to spontaneous polarization, and how this polarization can be switched.
- Understand how this polarization leads to the interesting electrical properties displayed by ferroelectrics.
- Understand how these properties are made use of in today's technologies.

Ferroelectricity

📖: Ferroelectricity is an electrical phenomenon whereby certain materials may exhibit a **spontaneous dipole moment**, the direction of which can be switched between equivalent states by the application of an external electric field.



A net permanent dipole moment from the vector sum of dipole moments in each unit cell, $\Sigma \mu$, cannot exist in a structure that has a center of symmetry, **why?**

As any dipole moment generated in one direction would be forced by symmetry to be zero. Therefore, ferroelectrics must be non-centrosymmetric. The central atom must be in a non-equilibrium position.

Ferroelectricity

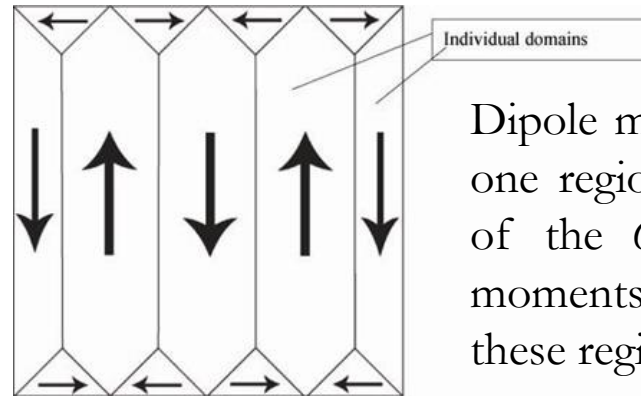
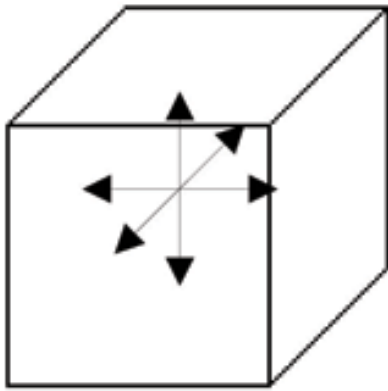
- ✦ The internal electric dipoles of a ferroelectric material are physically tied to the material lattice so anything that changes **the lattice structure** will change the strength of the dipoles and cause a current to flow into or out of the capacitor even without the presence of an external voltage across the capacitor.
- ✦ Two stimuli that will change the lattice dimensions of a material are **force and temperature**.
- ✦ The generation of a current in response to the application of a force to a capacitor is called **piezoelectricity**.
- ✦ The generation of current in response to a change in temperature is called **pyroelectricity**.

Ferroelectricity

- ✦ Placing a ferroelectric material between two conductive plates creates a **ferroelectric capacitor**.
- ✦ Ferroelectric capacitors exhibit **nonlinear** properties and usually have **very high dielectric constants**.
- ✦ The fact that the internal electric dipoles can be forced to change their direction by the application of an external voltage gives rise to **hysteresis** in the "**polarization vs voltage**" property of the capacitor.
- ✦ Polarization is defined as the total charge stored on the plates of the capacitor divided by the area of the plates.
- ✦ Hysteresis means memory and ferroelectric capacitors are used to make ferroelectric RAM for computers and RFID cards.

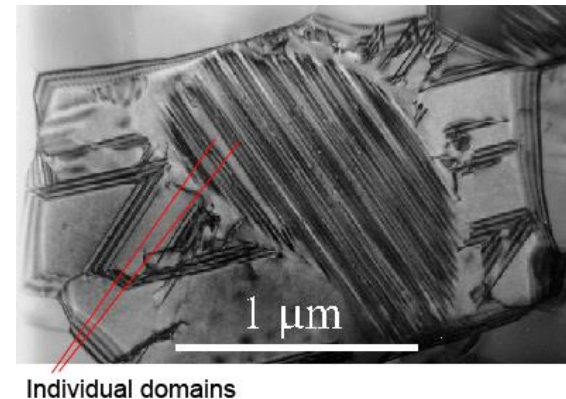
Domain Structure

Polarization may be defined as the total dipole moment per unit volume, i.e. $P = \frac{\sum \mu}{V}$



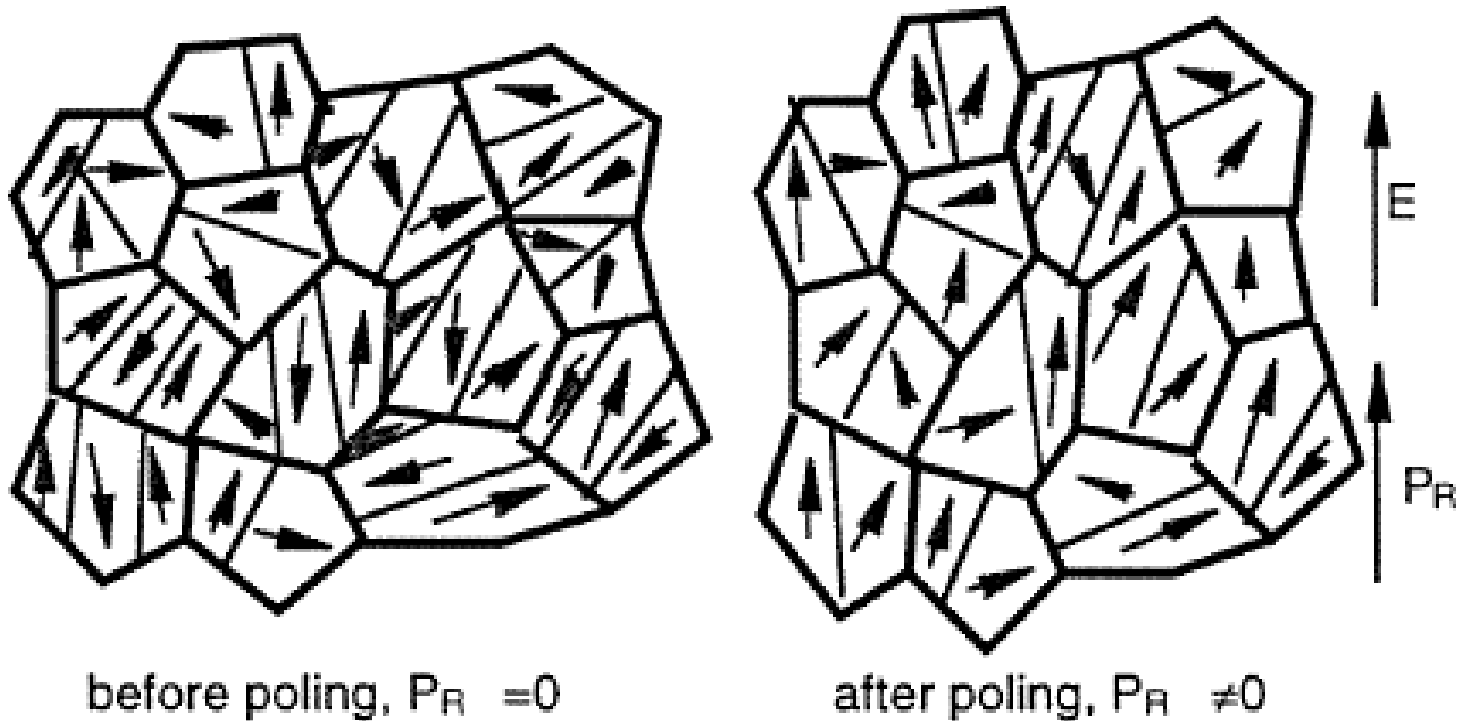
Dipole moments of the unit cells in one region lie along a different one of the 6 directions to the dipole moments in another region. Each of these regions is called a **domain**.

Polarized along a **unique crystallographic direction**, in that certain atoms are displaced along this axis, leading to a dipole moment along it.



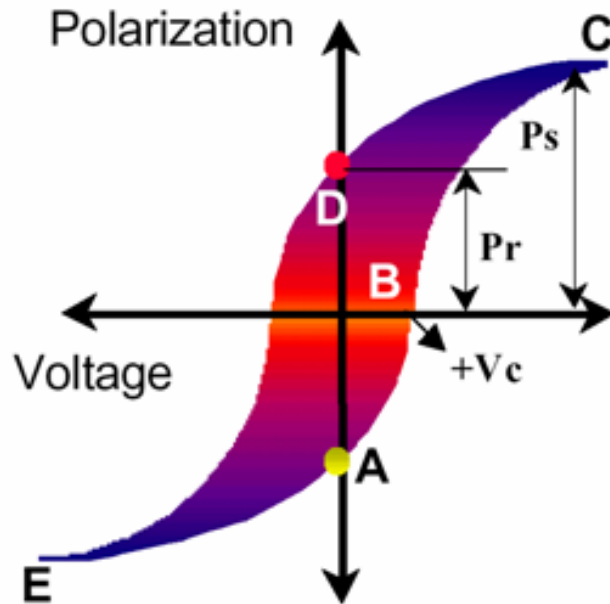
 : A **domain** is a homogenous region of a ferroelectric, in which all of the dipole moments in adjacent unit cells have the same orientation.

Poling of a polycrystalline ferroelectric

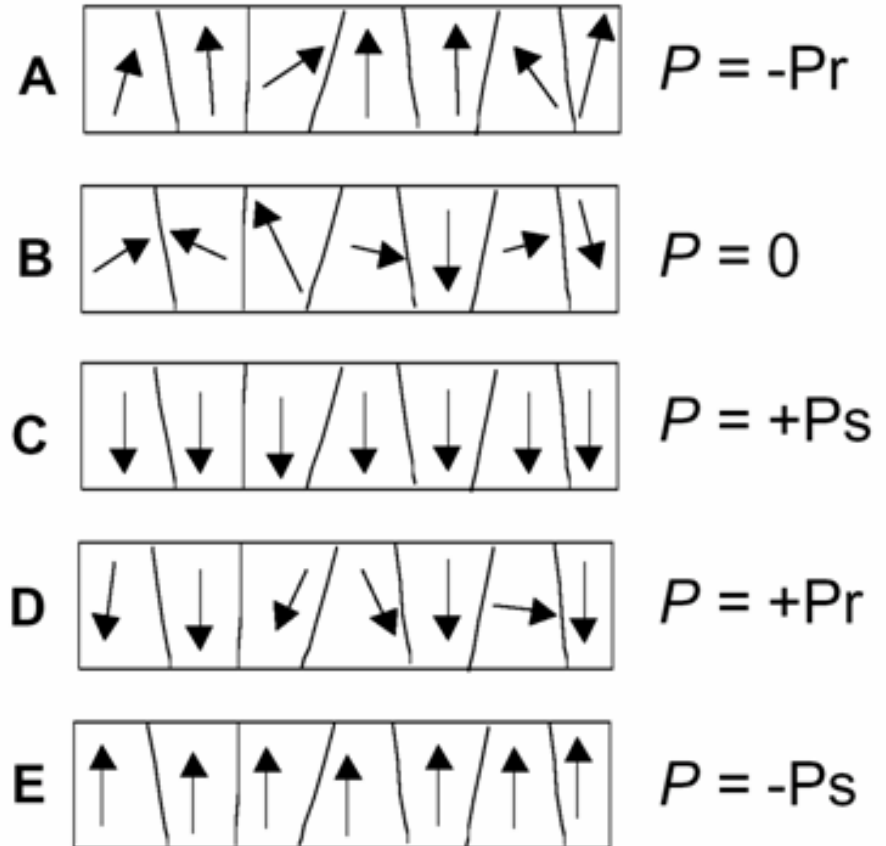


A polycrystalline ferroelectric with random orientation of grains before and after poling. Many domain walls are present in the poled material, however, the net remnant polarization is nonzero.

Domain Wall Movement

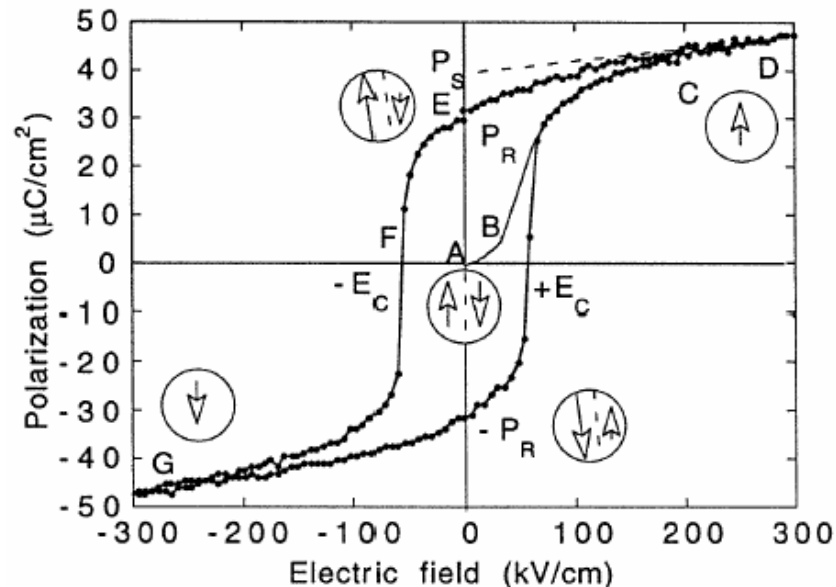


Domain movement



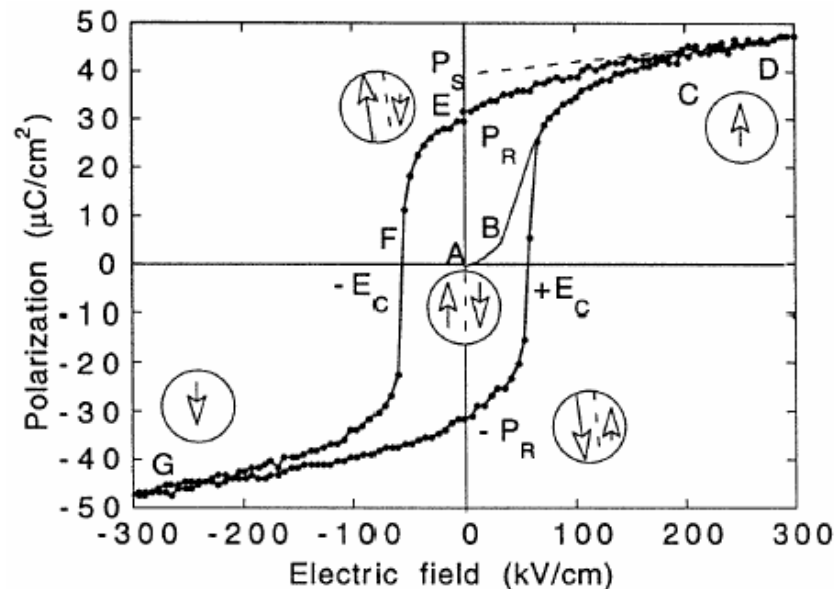
Ferroelectric hysteresis loop

- ⚡ The most important characteristic of ferroelectric materials is **polarization reversal** (or switching) by an electric field.
- ⚡ One consequence of the domain-wall switching in ferroelectric materials is the occurrence of the **ferroelectric hysteresis loop**.



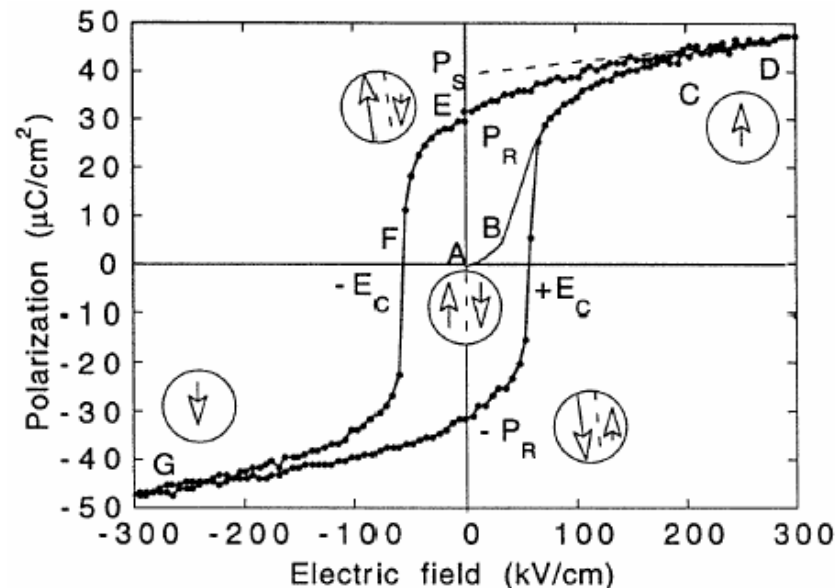
Ferroelectric hysteresis loop

- As the field is increased the polarization of domains with an unfavorable direction of polarization will start to switch in the direction of the field, rapidly increasing the measured charge density (segment BC).



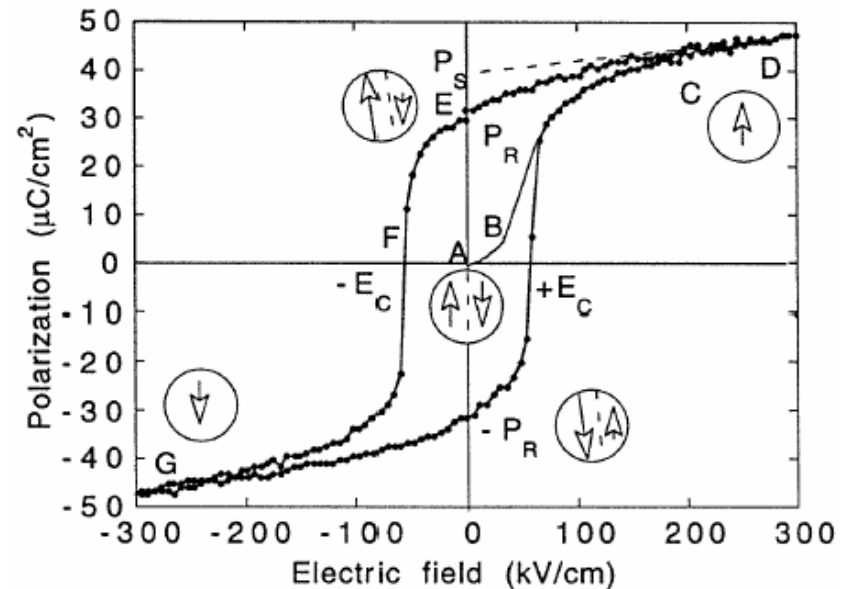
Ferroelectric hysteresis loop

- 1 The polarization response in this region is strongly nonlinear.
- 2 Once all the domains are aligned (point C) the ferroelectricity again behaves linearly (segment CD).
- 3 If the field strength starts to decrease, some domains will back-switch, **but at zero field the polarization is nonzero** (point E).
- 4 The value of polarization at zero field (point E) is called the **remnant polarization, P_R** .

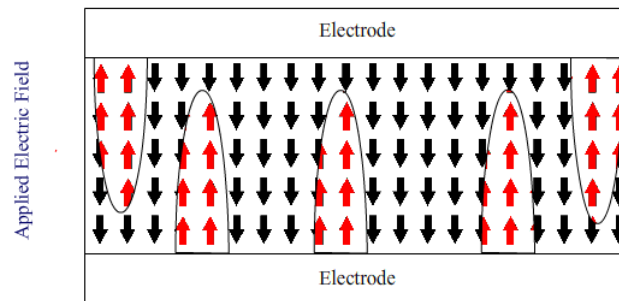
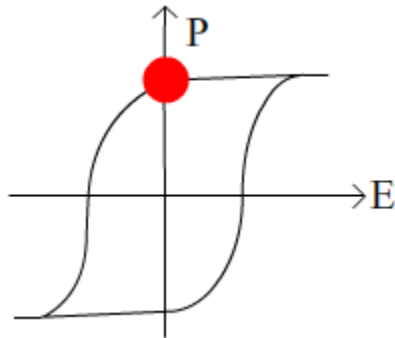
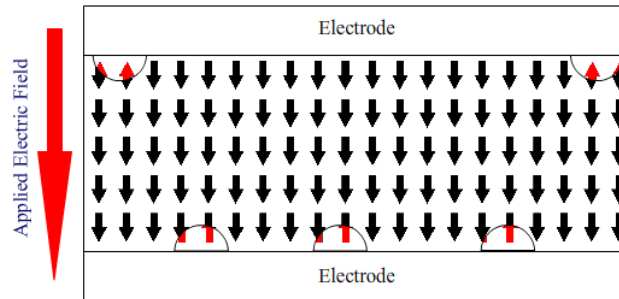
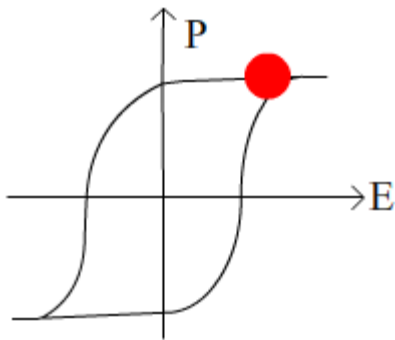
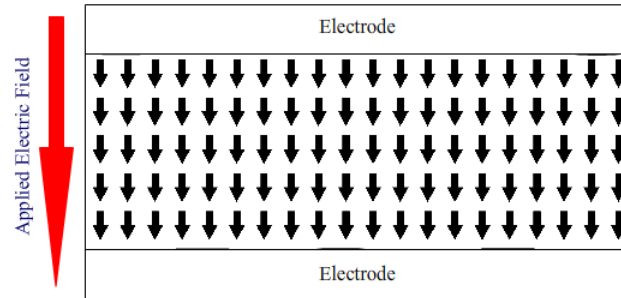
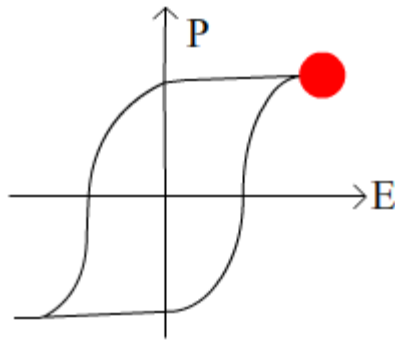


Ferroelectric hysteresis loop

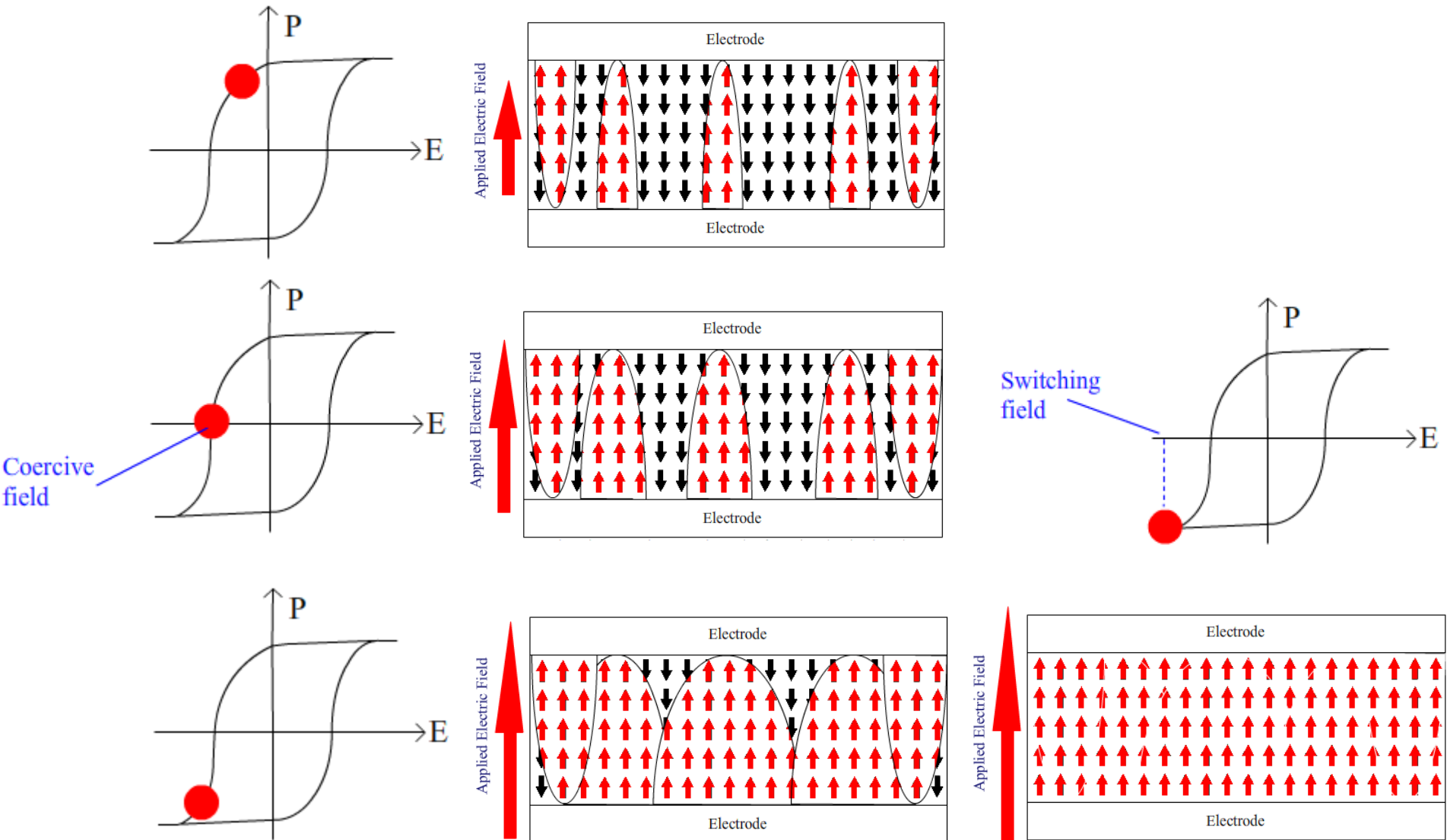
- ❧ To reach a zero polarization state the field must be reversed (point F).
- ❧ The field necessary to bring the polarization to zero is called the **coercive field, E_C** .
- ❧ The **spontaneous polarization P_S** is usually taken as the intercept of the polarization axis with the extrapolated linear segment CD.
- ❧ Further increase of the field in the negative direction will cause **a new alignment** of dipoles and saturation (point G).



Switching polarization

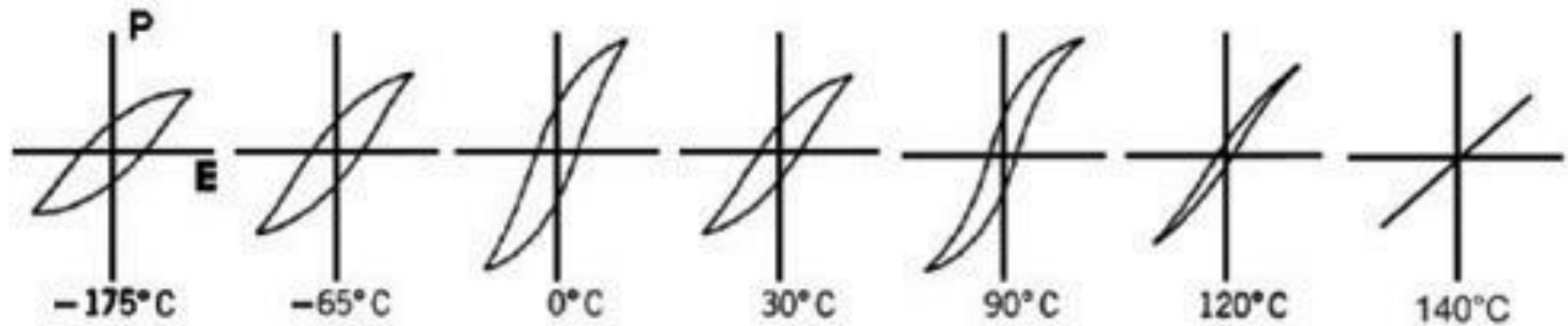


Switching polarization



Temperature dependence of the Hysteresis loop

What happens if the temperature is raised ?



Why should this be?

This is a general behavior that can arise for two reasons depending on the material.

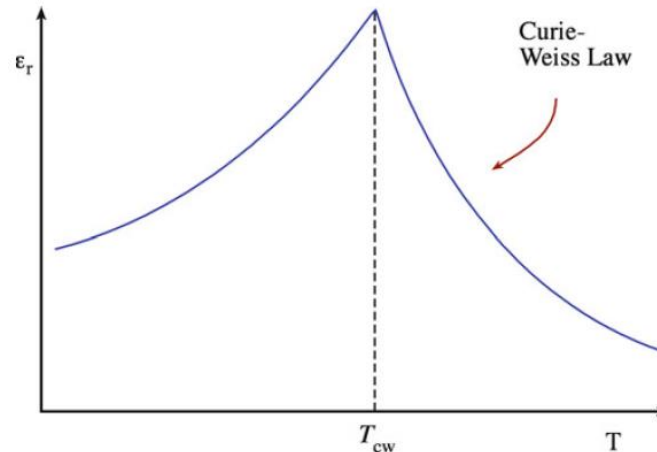
1) Disorder. Each unit cell has its own dipole moment, which, when there is a net polarisation, are described as ordered. At high T , the direction of the dipole moments randomises, giving a disordered material with no net polarization.

2) Phase transitions that can open up new possibilities for dipole moments to form. In this case, there is a jump at 0°C , and at 90°C , where the loop becomes taller.

Curie-Weiss law

- ✦ The temperature at which the spontaneous polarization disappears is called the **Curie temperature**, T_C .
- ✦ Above the Curie point the dielectric permittivity falls off with temperature according to the **Curie–Weiss law**

$$\varepsilon = \varepsilon_0 + \frac{C}{T - T_0} \approx \frac{C}{T - T_0}$$



📄 where C is the Curie constant, T_0 ($T_0 \leq T_C$) is the Curie–Weiss temperature.

Curie Point & Phase Transitions

- ✦ All ferroelectric materials have a transition temperature called the Curie point (T_c).
- ✦ At $T > T_c$ the crystal does not exhibit ferroelectricity, while for $T < T_c$ it is ferroelectric.
- ✦ On decreasing the temperature through the Curie point, a ferroelectric crystal undergoes a phase transition from a non-ferroelectric phase to a ferroelectric phase.
- ✦ If there are **more than one ferroelectric phases**, the T at which the crystal transforms from one phase to another is called the transition temperature.

Curie Point & Phase Transitions

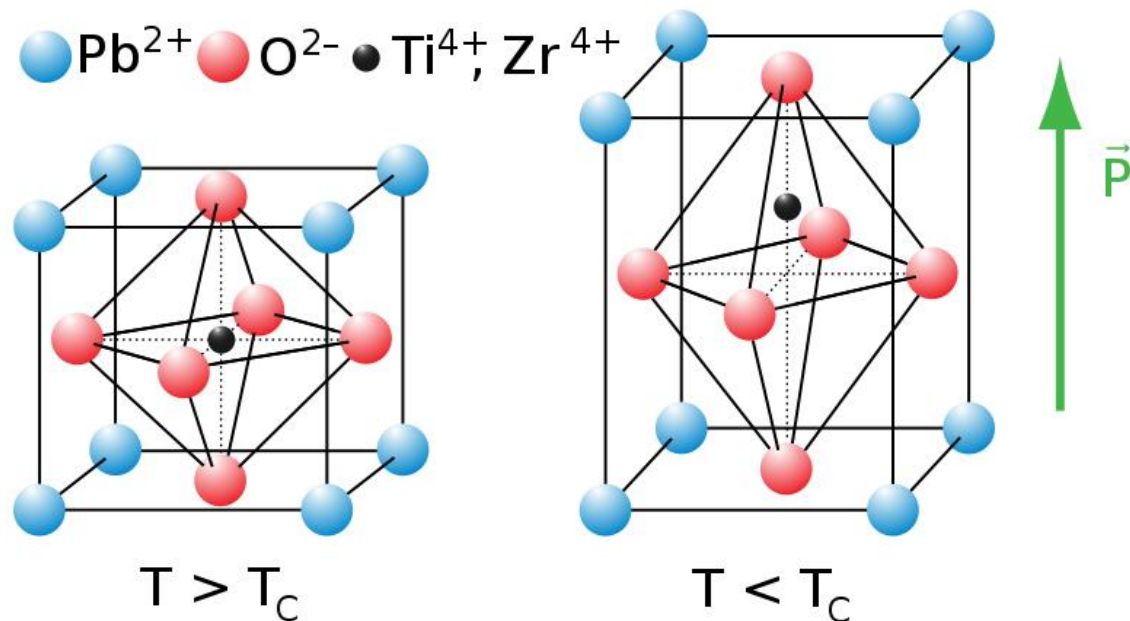
- ✦ For example, the variation of the relative permittivity ϵ_r with temperature as a BaTiO_3 crystal is cooled from its paraelectric **cubic phase** to the ferroelectric **tetragonal**, **orthorhombic**, and **rhombohedral phases**.
- ✦ Near the Curie point or transition temperatures, thermodynamic properties including dielectric, elastic, optical, and thermal constants show an anomalous behavior.
- ✦ This is due to a distortion in the crystal as the phase structure changes.

Phase Transformation

- ✦ Most ferroelectric materials undergo a structural phase transition from a high-temperature nonferroelectric (or paraelectric) phase into a low-temperature ferroelectric phase.
- ✦ The paraelectric phase may be piezoelectric or nonpiezoelectric and is rarely polar.
- ✦ The symmetry of the ferroelectric phase is always lower than the symmetry of the paraelectric phase.

Perovskites

- Perovskite is a family name of a group of materials and the mineral name of calcium titanate (CaTiO_3) having a structure of the type ABO_3 .
- Many piezoelectric (including ferroelectric) ceramics such as Barium Titanate (BaTiO_3), Lead Titanate (PbTiO_3), Lead Zirconate Titanate (PZT), Lead Lanthanum Zirconate Titanate (PLZT), Lead Magnesium Niobate (PMN), Potassium Niobate (KNbO_3), Potassium Sodium Niobate ($\text{K}_x\text{Na}_{1-x}\text{NbO}_3$), and Potassium Tantalate Niobate ($\text{K}(\text{Ta}_x\text{Nb}_{1-x})\text{O}_3$) have a perovskite type structure.



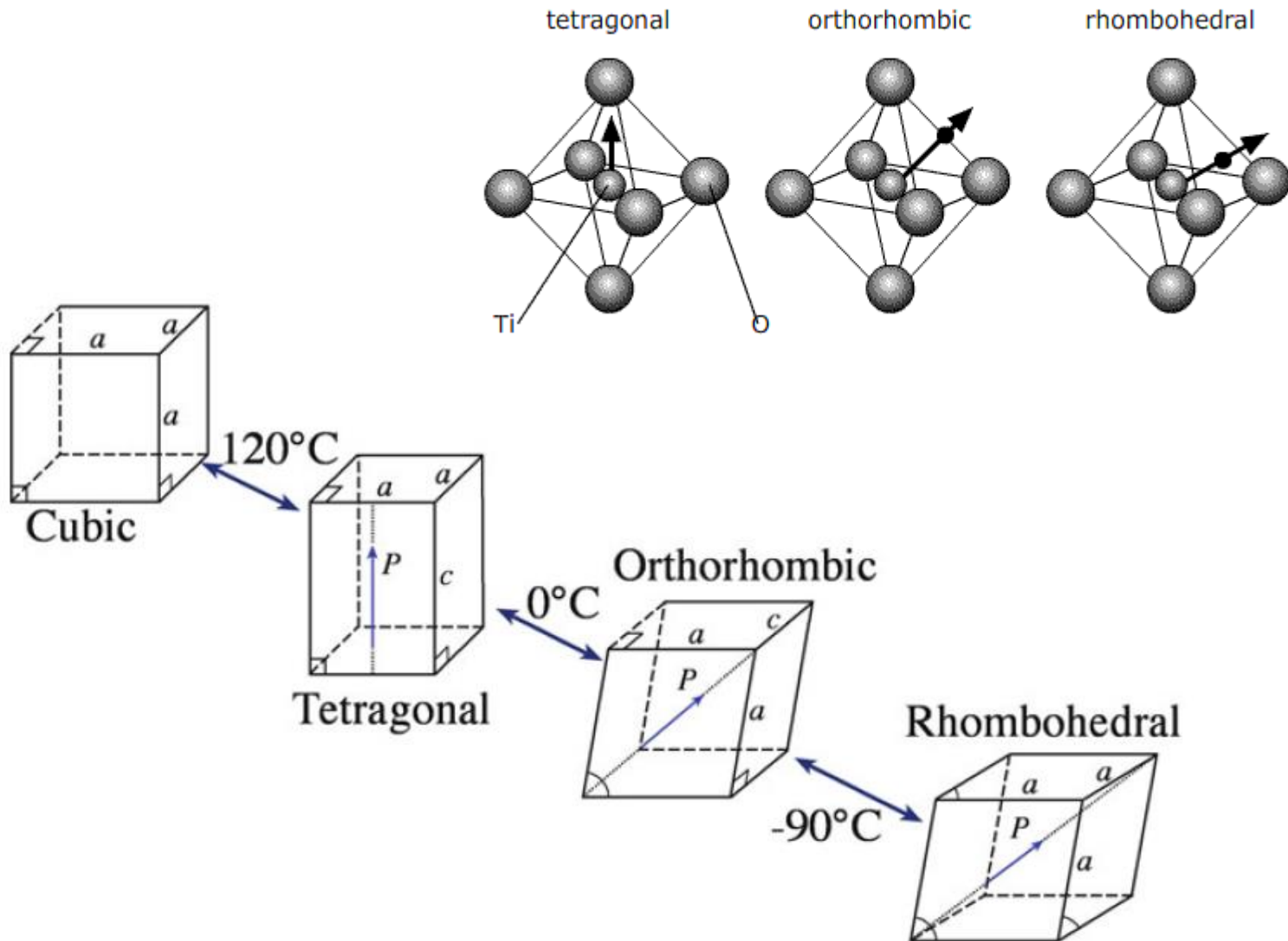
Size effect

- ✦ The dielectric properties of BaTiO_3 are found to be dependent on the grain size.
- ✦ Large grained BaTiO_3 shows an extremely high dielectric constant at the Curie point.
- ✦ This is because of the formation of multiple domains in a single grain, the motion of whose walls increases the dielectric constant at the Curie point.
- ✦ For a BaTiO_3 ceramic with fine grains, a single domain forms inside each grain.
- ✦ The movement of domain walls are restricted by the grain boundaries, thus leading to a low dielectric constant at the Curie point as compared to coarse grained BaTiO_3 .

PLZT

- ✦ The electro-optic applications of PLZT ceramics depends on the composition.
- ✦ PLZT ceramic compositions in the **tetragonal ferroelectric** (F_T) region show hysteresis loops with a very high coercive field (E_C).
- ✦ Materials with this composition exhibit linear electro-optic behavior for $E < E_C$.
- ✦ PLZT ceramic compositions in the **rhombohedral ferroelectric** (F_R) region of the PLZT phase diagram have loops with a low coercive field.
- ✦ These PLZT ceramics are useful for optical memory applications.

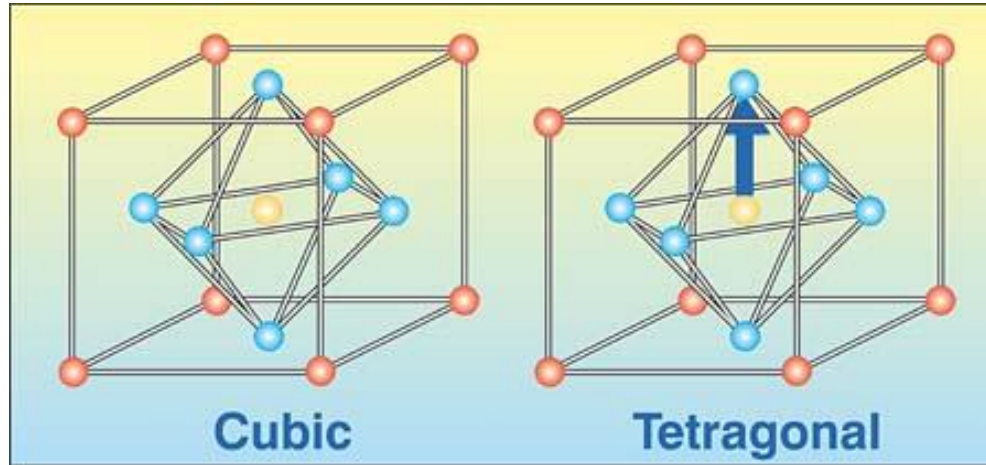
BaTiO₃ and phase transformation



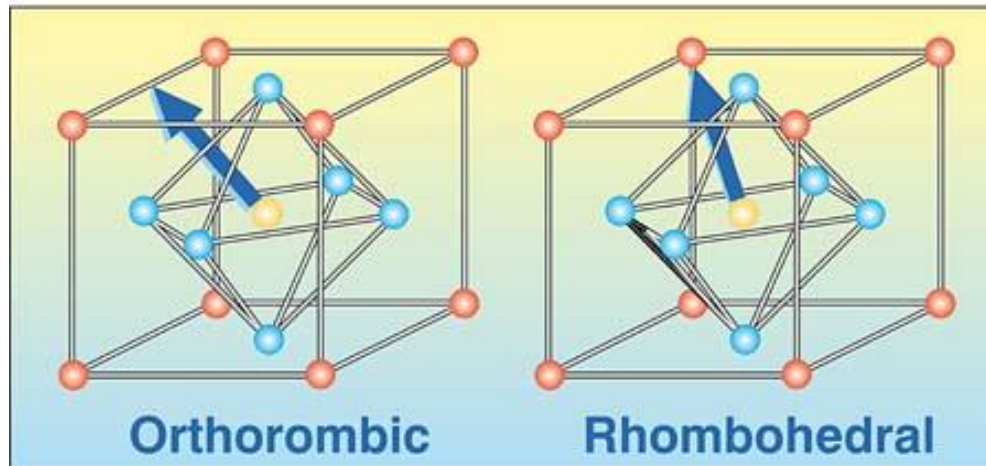
BaTiO₃ and phase transformation

- ✦ BaTiO₃ has a paraelectric *cubic phase* above its Curie point of about 120°C.
- ✦ In the T of 120°C to 0°C, the ferroelectric *tetragonal phase* with a c/a ratio of ~ 1.01 is stable.
- ✦ The spontaneous polarization is along one of the [001] directions in the original cubic structure.
- ✦ Between 0°C and -90°C, the ferroelectric *orthorhombic phase* is stable with the polarization along one of the [110] directions in the original cubic structure.
- ✦ On decreasing T below -90°C the phase transition from the orthorhombic to ferroelectric *rhombohedral phase* leads to polarization along one of the [111] cubic directions.

Perovskite and phase transformation



[001]
directions

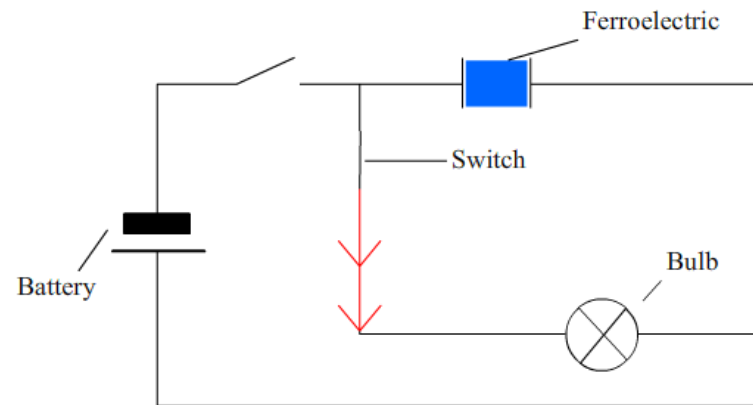
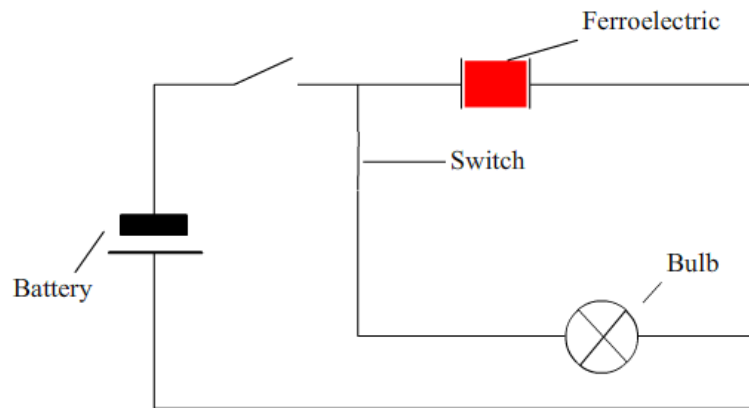
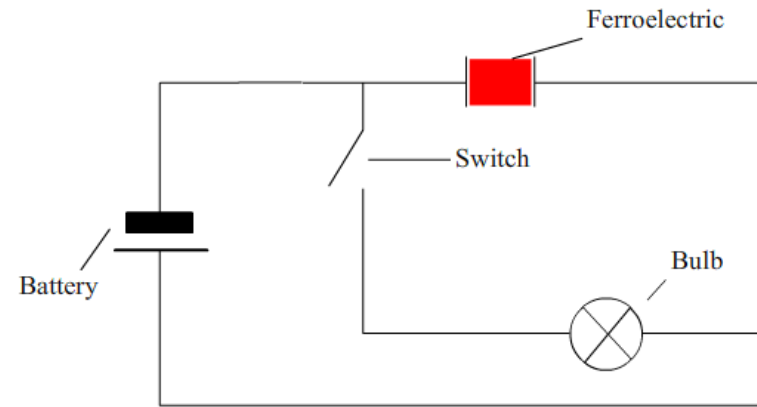
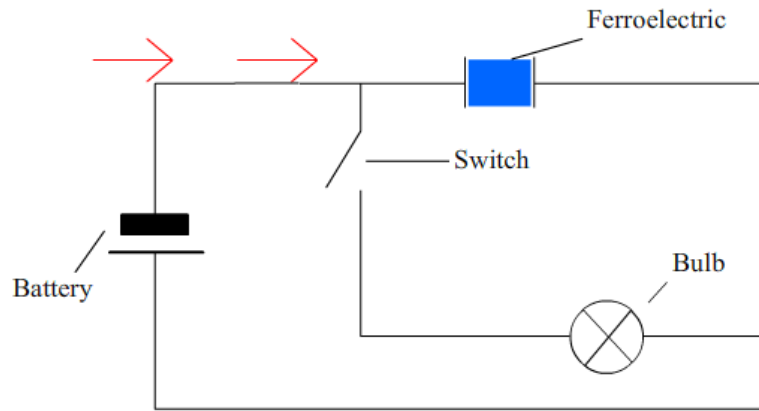


[110]
directions

[111]
directions

The phase transition sequence in perovskites

Ferroelectric- So what ?



Application of Ferroelectric Ceramics

- ✦ The combined properties of memory, piezoelectricity, and pyroelectricity make ferroelectric capacitors some of the most useful technological devices in modern society.
- ✦ Ferroelectric capacitors are at the heart of **medical ultrasound machines, high quality infrared cameras, fire sensors, sonar, vibration sensors, and even fuel injectors on diesel engines.**
- ✦ The high dielectric constants of ferroelectric materials used to concentrate large values of capacitance into small volumes, resulting in the very tiny surface mount capacitor.
- ✦ The electrooptic modulators that form the backbone of the Internet are made with ferroelectric materials.



Problems in Ferroelectric

- ✦ Problems associated with applications of ferroelectric materials, such as
 - 📄 polarization fatigue,
 - 📄 ageing and field and frequency dependence of the piezoelectric,
 - 📄 elastic and dielectric properties.
 - 📄 polycrystalline ferroelectrics and films: properties are often controlled by contributions from domain-wall displacements, extrinsic contributions, theoretical treatment presents a considerable challenge.
 - 📄 Geometry of thin films imposes boundary conditions which sometimes lead to very different properties of films with respect to bulk materials and which must be taken into account when modelling devices.

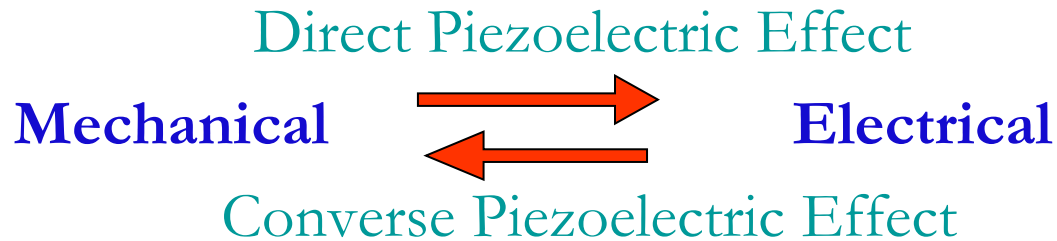
Part III Piezoelectric Properties

Aims:

- Understand the atomic basis for piezoelectricity.
- Understand how this basis scales up to a full effect on the macro scale.
- Understand how piezoelectrics tie in with ferroelectrics and how their properties arise from the same basis.
- Understand how the properties of piezoelectrics are put to use both industrially and commercially.

Piezoelectricity

Mechanical and electrical energy conversion phenomena, discovered by France Scientist **Pierre and Jacques Curie** brother in 1880.



When a piezoelectric is placed under a mechanical stress, the atomic structure of the crystal changes, such that ions in the structure separate, and a dipole moment is formed. For a net polarization to develop, the dipole formed must not be cancelled out by other dipoles in the unit cell. To do this, the piezoelectric atomic structure must be non-centrosymmetric.

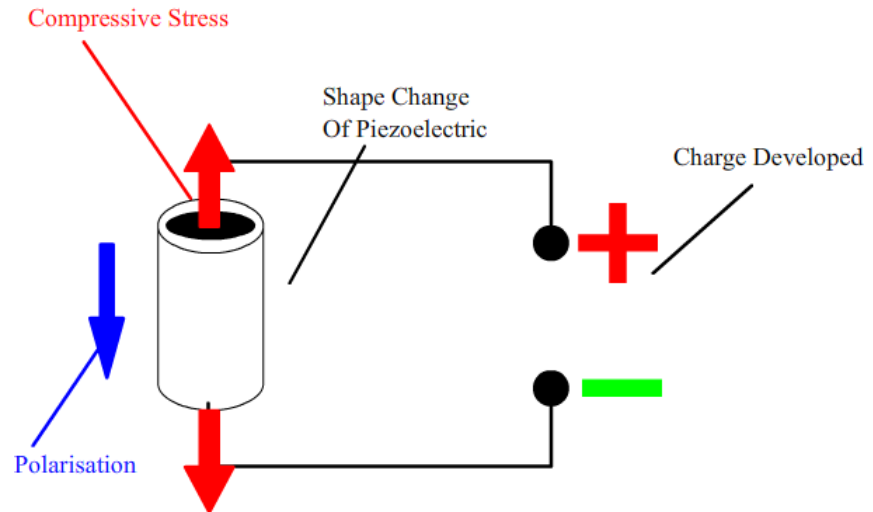
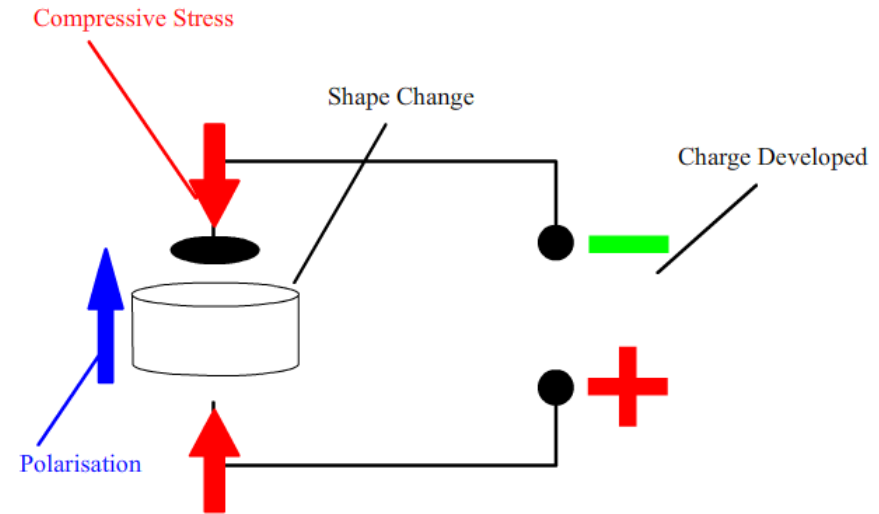
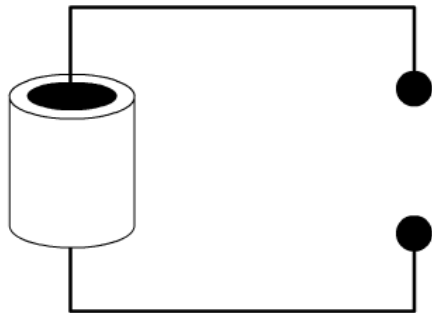
Piezoelectric Ceramics

The famous one is the lead zirconium titanium, $\text{Pb}(\text{Zr,Ti})\text{O}_3$ [PZT].

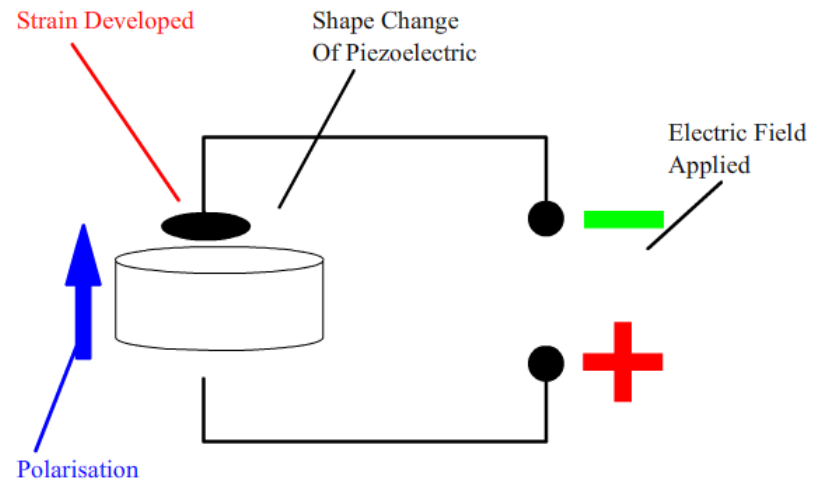
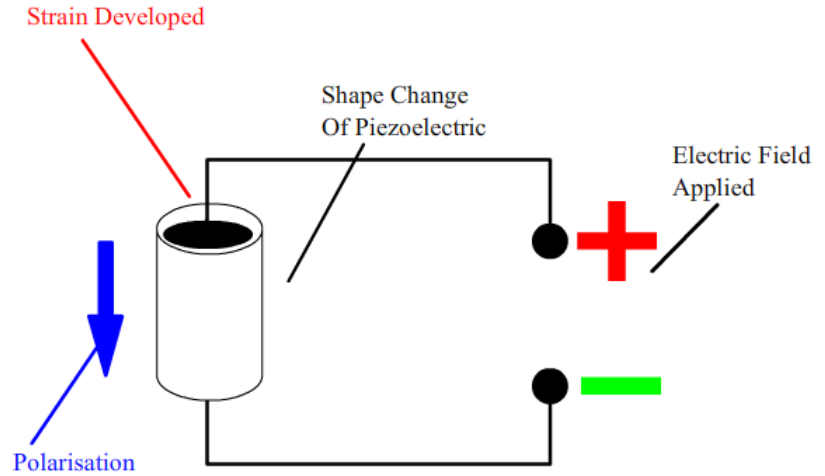
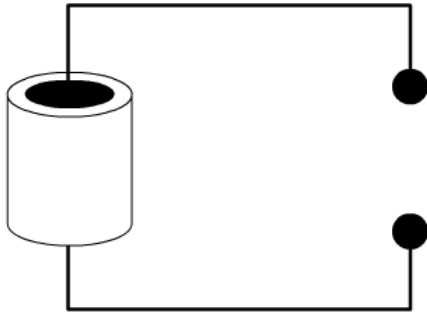
PZT was first reported as ferroelectric materials in 1950 by Shirane et al., and reported as a useful piezoelectric materials by Jaffe et al. in 1954.

Below the Curie temperature, PZT ceramics is ferroelectrics with piezoelectricity, and it was paraelectrics without piezoelectricity at the temperature higher than Curie Temperature.

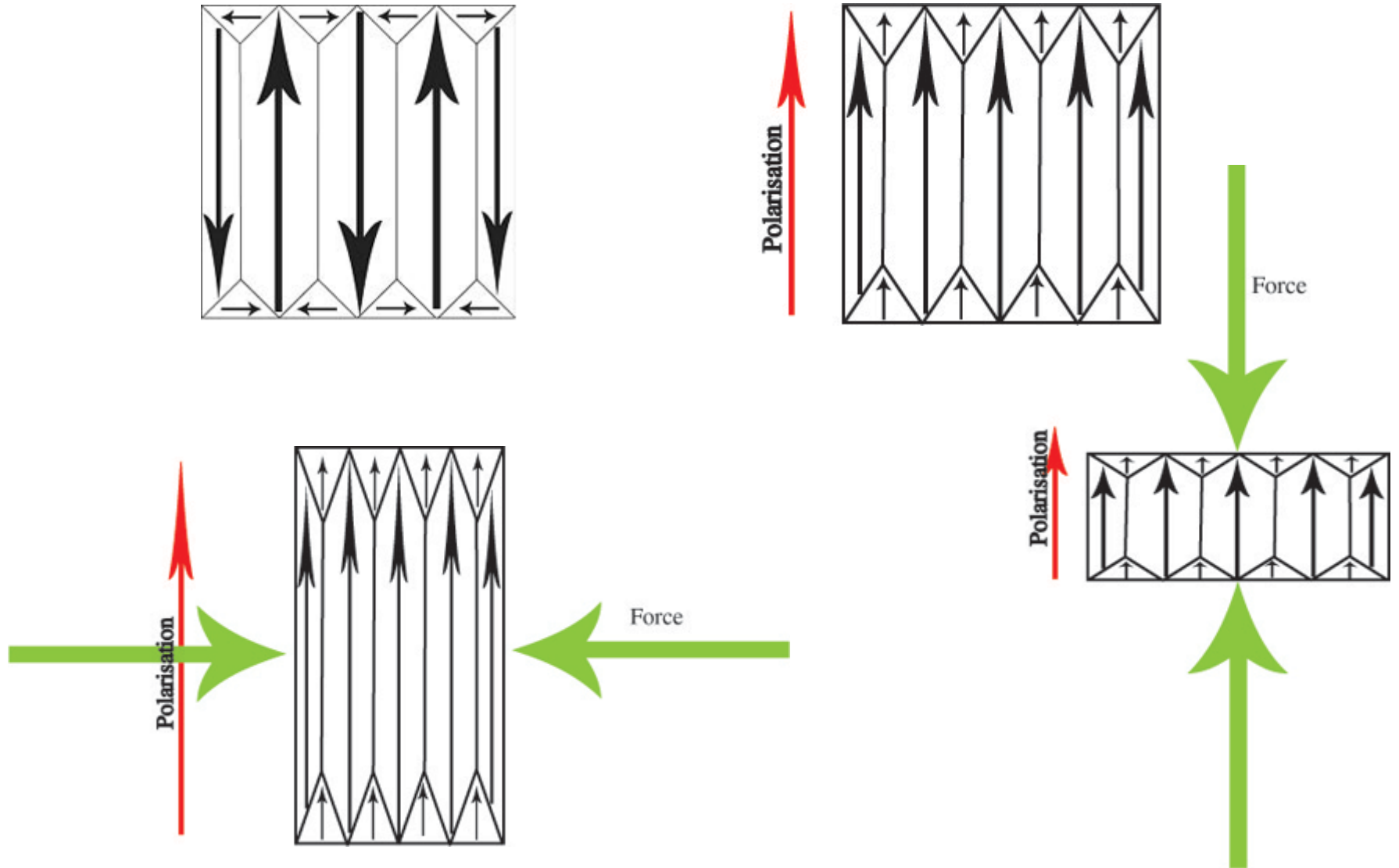
Piezoelectric effect



Piezoelectric effect



Spontaneously polarized piezoelectrics



Depolarization

1. Thermal depoling

If the material is exposed to excessive heat, such that its temperature approaches its Curie temperature, the dipole moments regain their unaligned state. At the Curie temperature, a ferroelectric becomes entirely unaligned. In order to prevent this occurring, it is sensible to use piezoelectrics well below their Curie temperature.

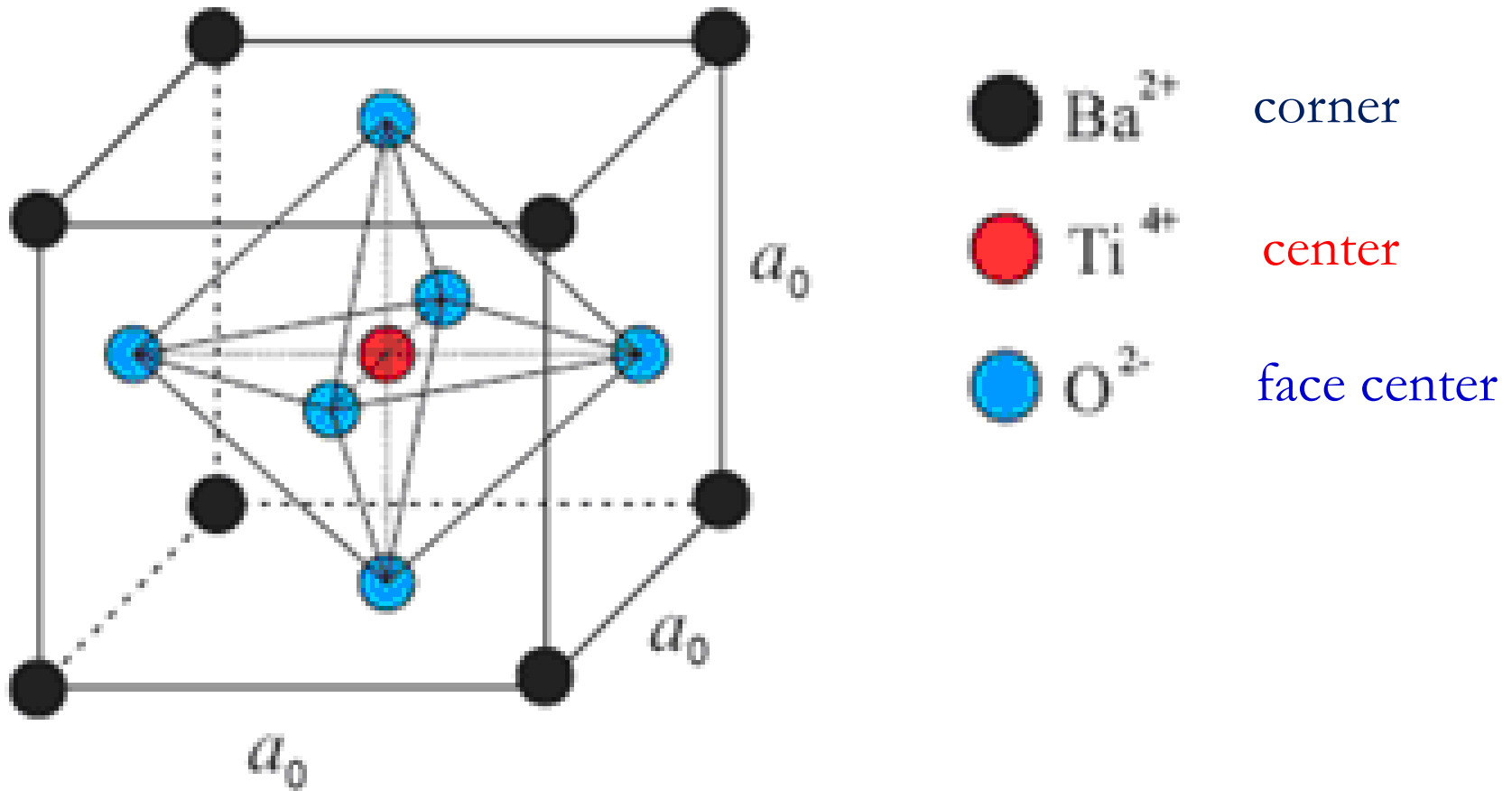
2. Electrical depoling

A strong electric field, when applied in the reverse direction to the already poled material, will lead to depoling. If an alternating field is used to produce ultrasound waves (see later) the field will depolarise the piezoelectric during the periods in which it is opposing the polarisation.

3. Mechanical depoling

If the stress placed on a piezoelectric is too high, it is possible to immediately depolarise the piezoelectric as the atom positions are altered. This completely ruins its properties.

Piezoelectric Ceramics

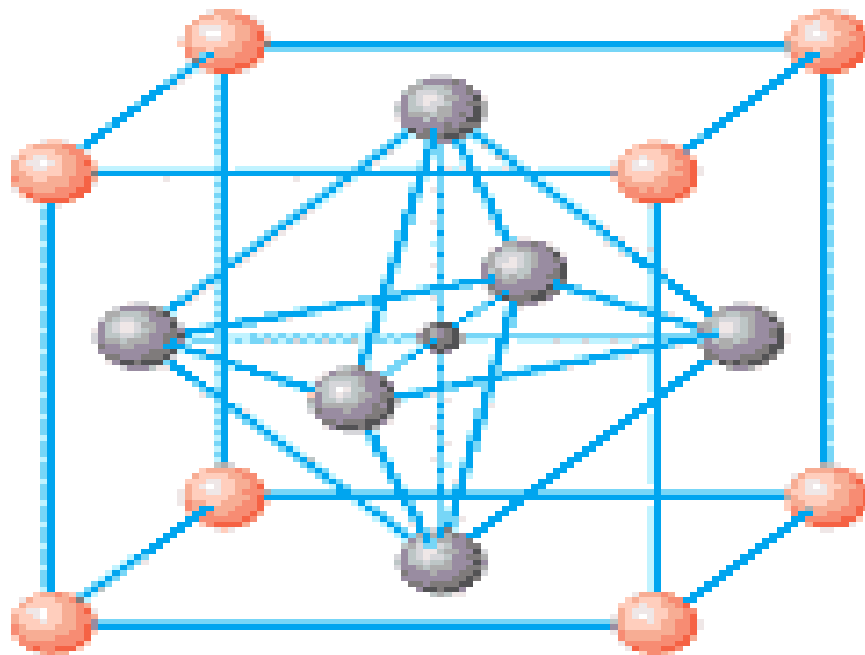


Pervoskite structure, with the chemical formula as ABO_3 e.g. : BaTiO_3

Piezoelectric Ceramics

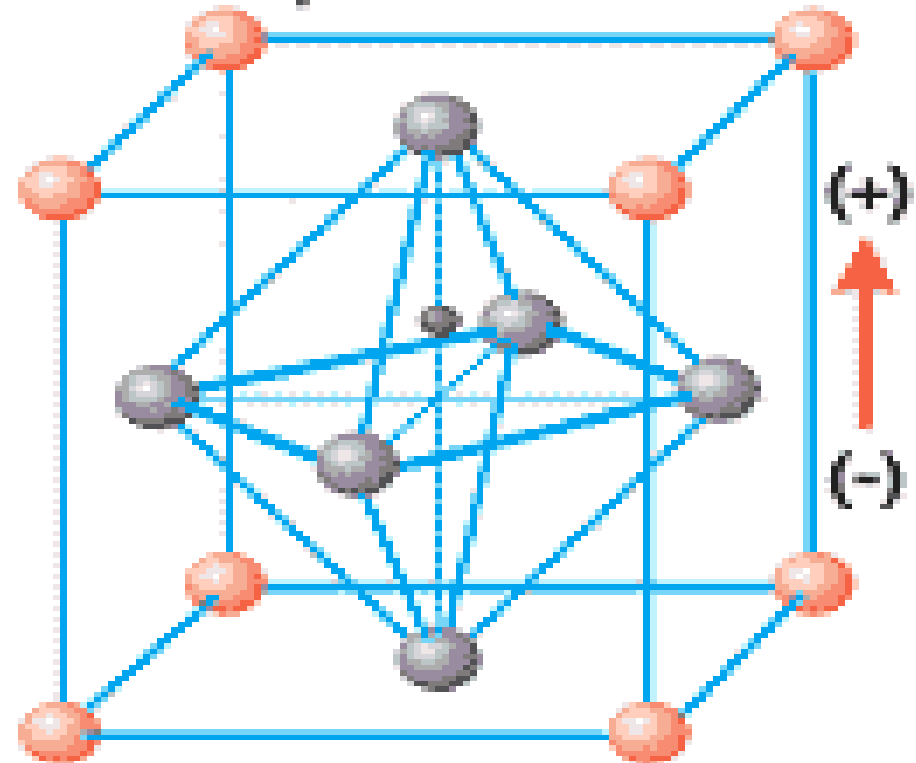
-  A^{2+} = Pb, Ba, other large, divalent metal ion
-  O^{2-} = oxygen
-  B^{4+} = Ti, Zr, other smaller, tetravalent metal ion

(a) temperatures above
Curie point



cubic lattice, symmetric
arrangement of positive and
negative charges

(b) temperatures below
Curie point

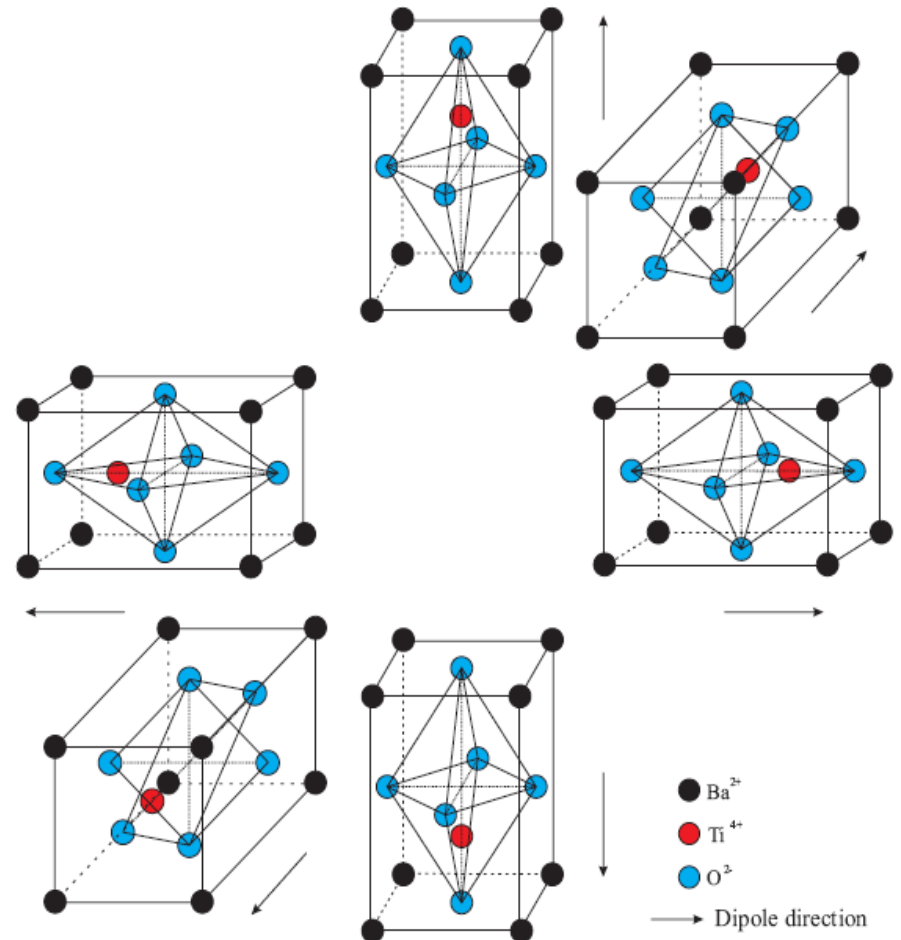


tetragonal (orthorhombic) lattice,
crystal has electric dipole

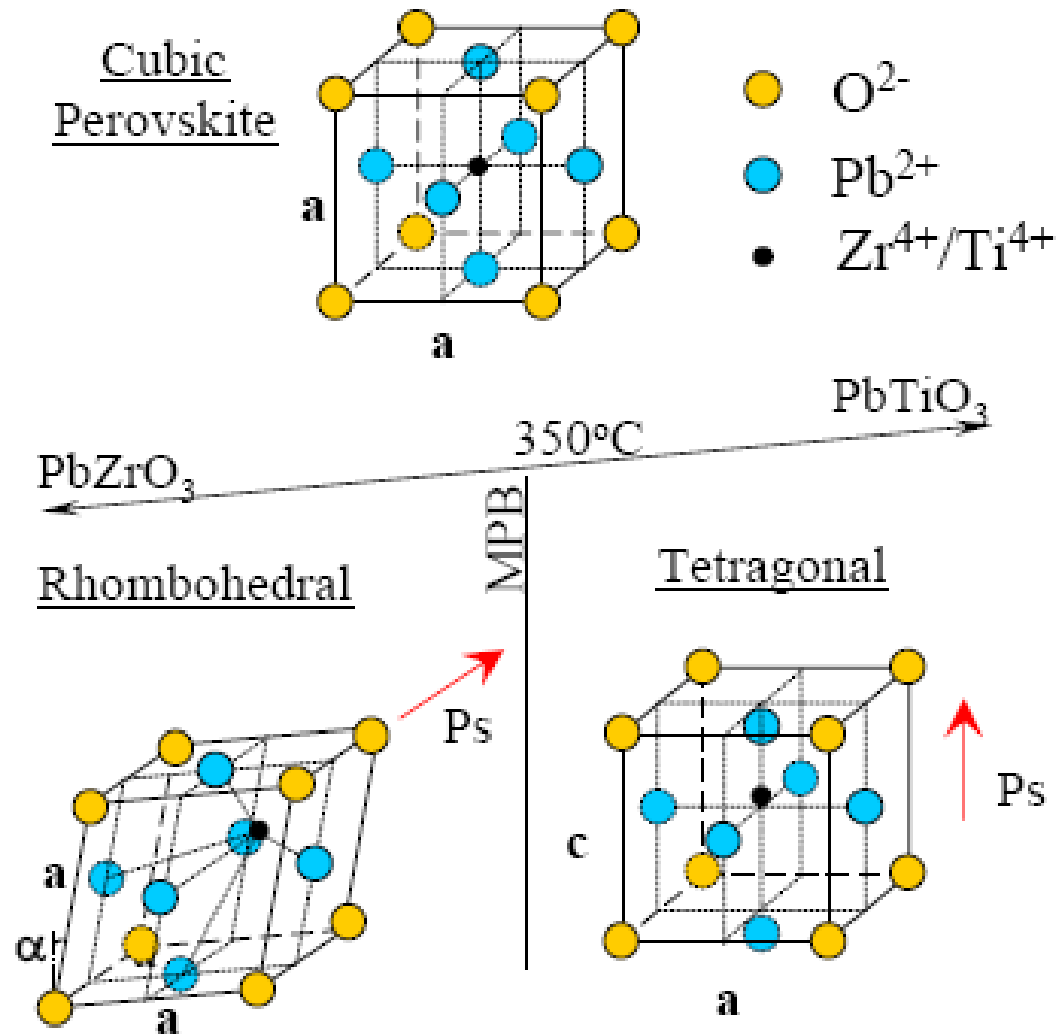
Piezoelectric Ceramics

☆ For a three-dimensional unit cell, there are six possible directions to displace the central positive ion towards the face centered negative ions.

☆ Therefore, there are six different choices for spontaneous polarization in the ferroelectric phase



Piezoelectric Ceramics



Piezoelectric effect

- ✦ The linear relationship between stress X_{jk} applied to a piezoelectric material and resulting charge density D_i is known as the **direct piezoelectric effect** and may be written as

$$D_i = d_{ijk} X_{jk}$$

- ✦ where d_{ijk} (C N^{-1}) is a third-rank tensor of piezoelectric coefficients.

Piezoelectric effect

- ✦ Another interesting property of piezoelectric material is they change their dimensions (contract or expand) when an electric field is applied to them.
- ✦ The **converse piezoelectric effect** describes the strain that is developed in a piezoelectric material due to the applied electric field:

$$x_{ij} = d_{kij} E_k = d_{ijk}^t E_k$$

- ✦ where t denotes the transposed matrix.
- ✦ The units of the converse piezoelectric coefficient are (m V⁻¹).

Piezoelectric effect

- ✎ The piezoelectric coefficients, d for the direct and converse piezoelectric effects are thermodynamically identical, i.e.

$$d_{\text{direct}} = d_{\text{converse}}$$

- ✎ Note that the sign of the piezoelectric charge D_i and strain ε_{ij} depends on the direction of the mechanical and electric fields, respectively.
- ✎ The piezoelectric coefficient d can be either positive or negative.

Piezoelectric effect

- ✦ It is common to call a piezoelectric coefficient measured in the direction of applied field the **longitudinal coefficient**, and that measured in the direction perpendicular to the field the **transverse coefficient**.
- ✦ Other piezoelectric coefficients are known as **shear coefficients**.
- ✦ Because the strain and stress are symmetrical tensors, the piezoelectric coefficient tensor is symmetrical with respect to the same indices, $d_{ijk} = d_{ikj}$.

Piezoelectricity

- ✦ The microscopic origin of the piezoelectric effect is the displacement of **ionic charges** within a crystal structure.
- ✦ In the absence of external strain, the charge distribution is symmetric and the net electric dipole moment is zero.
- ✦ However when an external stress is applied, the charges are displaced and the charge distribution is no longer symmetric and a net polarization is created.

Piezoelectricity

- ✪ In some cases a crystal possesses a unique polar axis even in the unstrained condition.
- ✪ This can result in a change of the electric charge due to a uniform change of temperature.
- ✪ This is called the *pyroelectric effect*.
- ✪ The direct piezoelectric effect is the basis for force, pressure, vibration and acceleration sensors and the converse effect for actuator and displacement devices.

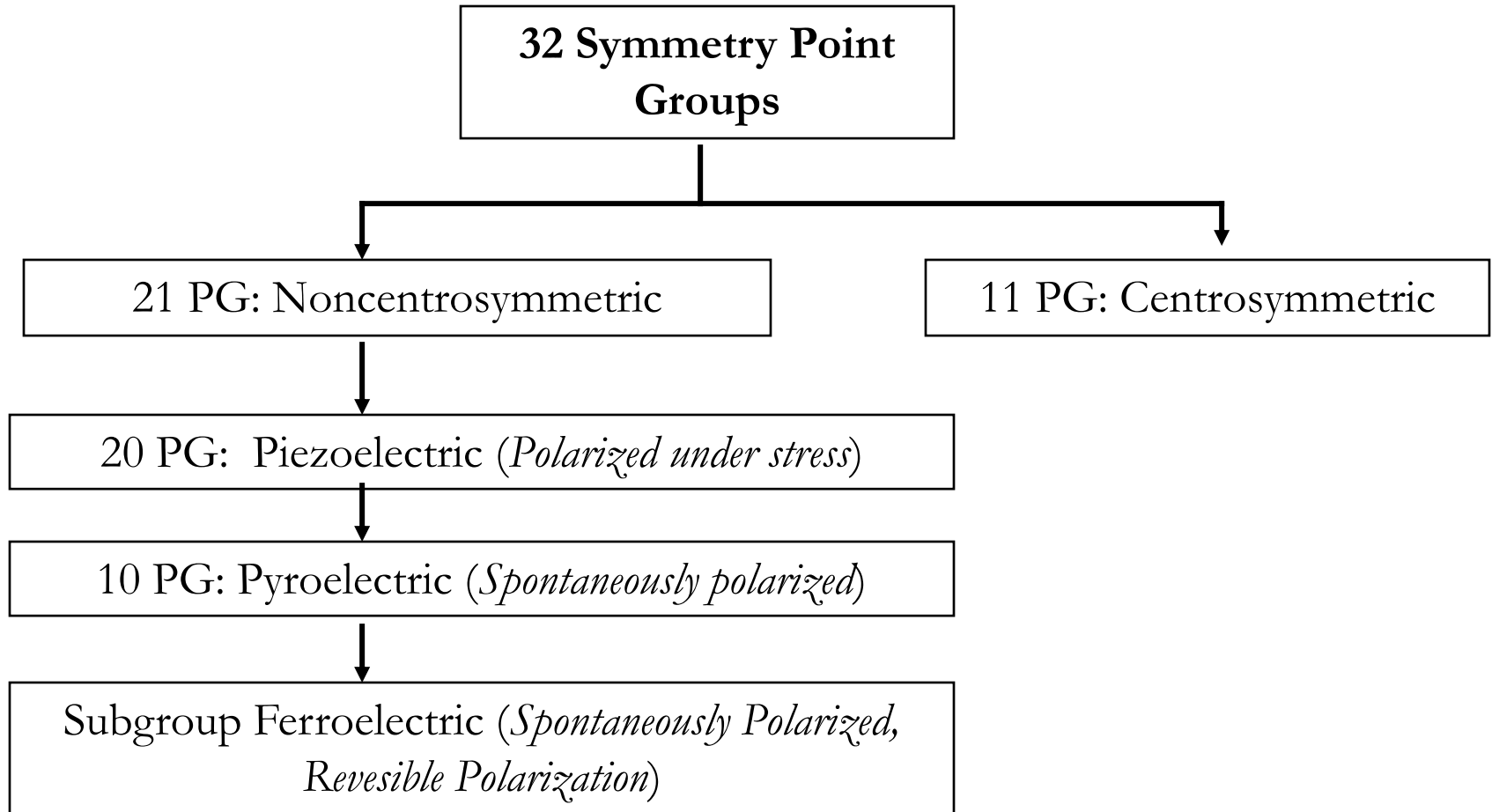
Piezoelectric and subgroup

- ✎ The elements of symmetry that are utilized by crystallographers to define symmetry about a point in space, for example, the central point of unit cell, are
 - 📄 a point (center) of symmetry,
 - 📄 axes of rotation,
 - 📄 mirror planes, and
 - 📄 combinations of these.
- ✎ Utilizing these symmetry elements, all crystals can be divided into **32 different classes or point groups**.

Piezoelectric and subgroup

- ✦ These 32 point groups are subdivisions of 7 basic crystal systems:
 - 📄 **triclinic,**
 - 📄 **monoclinic,**
 - 📄 **orthorhombic,**
 - 📄 **tetragonal,**
 - 📄 **rhombohedral (trigonal),**
 - 📄 **hexagonal, and**
 - 📄 **cubic.**
- ✦ Of the 32 point groups, 21 classes do not possess a center of symmetry (*a necessary condition for piezoelectricity to exist*) and 20 of these are piezoelectric.
- ✦ One class, although lacking a center of symmetry, is not piezoelectric because of other combined symmetry elements.

Piezoelectric and subgroup



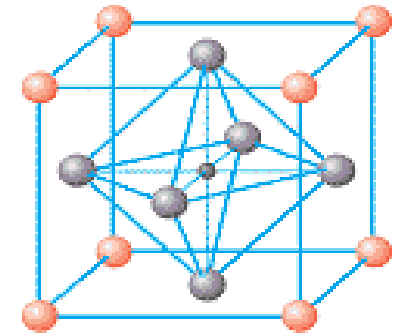
Piezoelectric and subgroup

- ✦ As discussed in previously slide, piezoelectric coefficients must be zero and the piezoelectric effect is absent in all 11 centrosymmetric point groups.
- ✦ Materials that belong to other symmetries may exhibit the piezoelectric effect.

How are piezoelectric ceramics made?

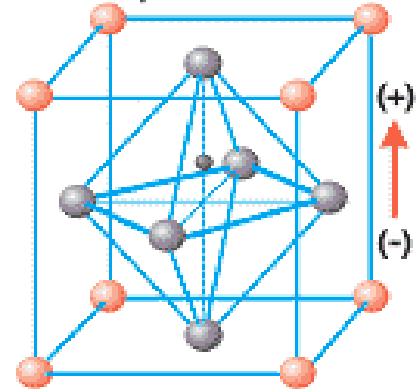
- ✦ A traditional piezoelectric ceramic is **perovskite crystal**, each consisting of a small, tetravalent metal ion, usually titanium or zirconium, in a lattice of larger, divalent metal ions, usually lead or barium, and O^{2-} ions.
- ✦ Under conditions that confer tetragonal or rhombohedral symmetry on the crystals, each crystal has a dipole moment.

(a) temperatures above Curie point



cubic lattice, symmetric arrangement of positive and negative charges

(b) temperatures below Curie point



tetragonal (orthorhombic) lattice, crystal has electric dipole

- A^{2+} = Pb, Ba, other large, divalent metal ion
- O^{2-} = oxygen
- B^{4+} = Ti, Zr, other smaller, tetravalent metal ion

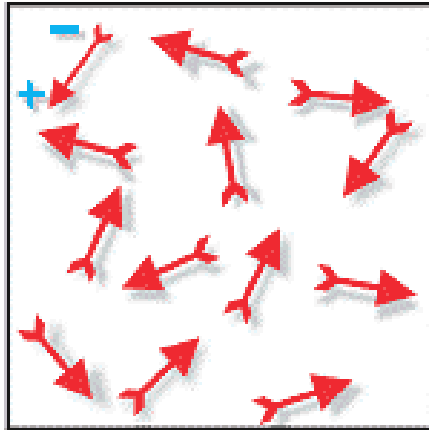
Polarization of piezoelectric

- ✦ Above a critical temperature, the *Curie point*, each perovskite crystal exhibits a simple cubic symmetry with no dipole moment.
- ✦
- ✦ At temperatures below the Curie point, however, each crystal has tetragonal or rhombohedral symmetry and a dipole moment.
- ✦
- ✦ Adjoining dipoles form regions of local alignment called *domains*.
- ✦
- ✦ The alignment gives a net dipole moment to the domain, and thus a net polarization.
- ✦ The direction of polarization among neighboring domains is random, however, so the ceramic element has no overall polarization.

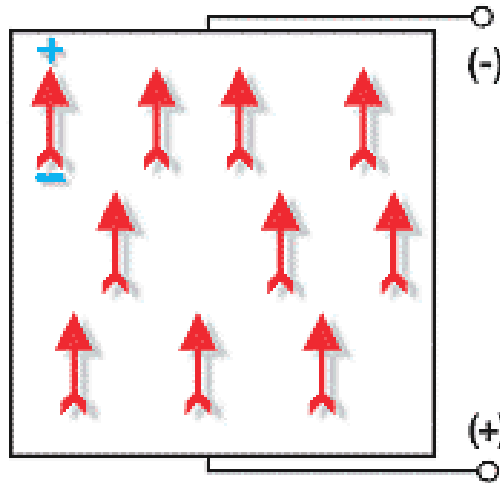
Polarization of piezoelectric

- ✪ The domains in a ceramic element are aligned by exposing the element to a strong, direct current electric field, usually at a temperature slightly below the Curie point.
- ✪ Through this polarizing (*poling*) treatment, domains most nearly aligned with the electric field expand at the expense of domains that are not aligned with the field, and the element lengthens in the direction of the field.
- ✪ When the electric field is removed most of the dipoles are locked into a configuration of near alignment.
- ✪ The element now has a permanent polarization, the remanent polarization, and is permanently elongated.

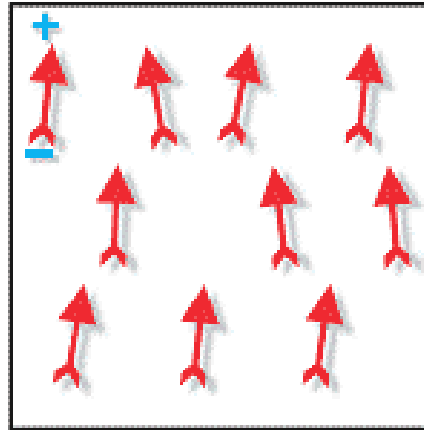
(a) random orientation of polar domains prior to polarization



(b) polarization in DC electric field



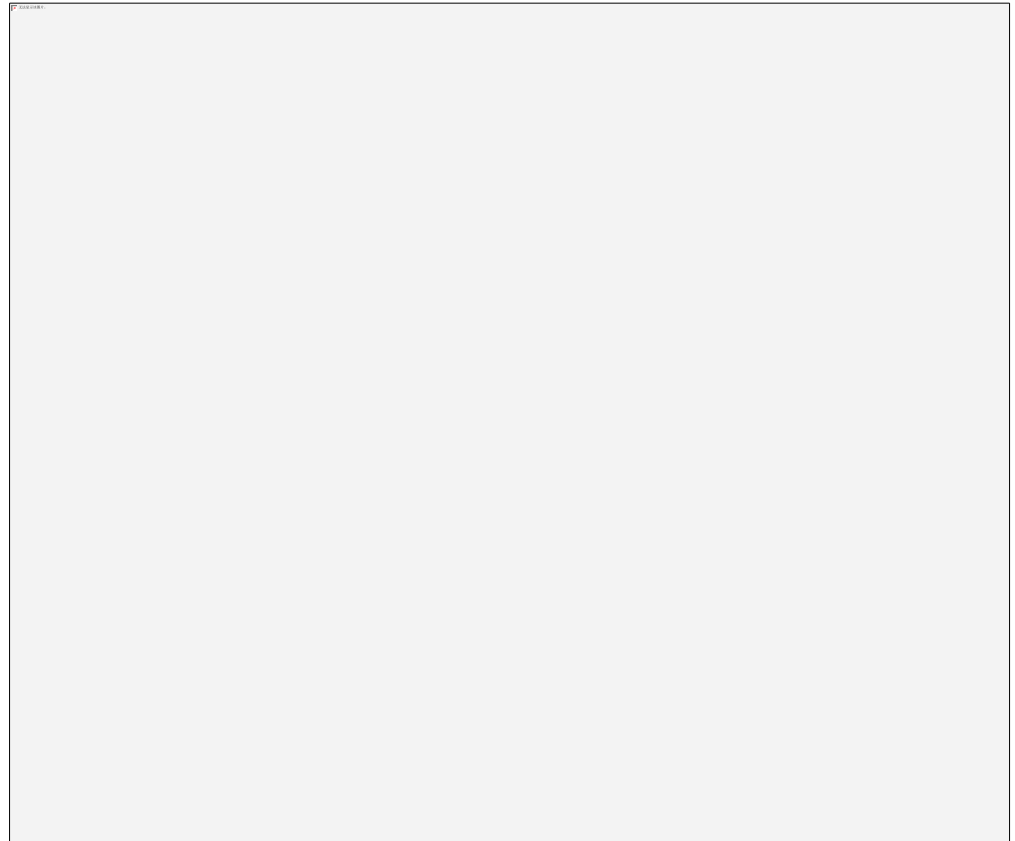
(c) remanent polarization after electric field removed



Electric dipoles in Weiss domains; (1) unpoled ferroelectric ceramic, (2) during and (3) after poling (piezoelectric ceramic)

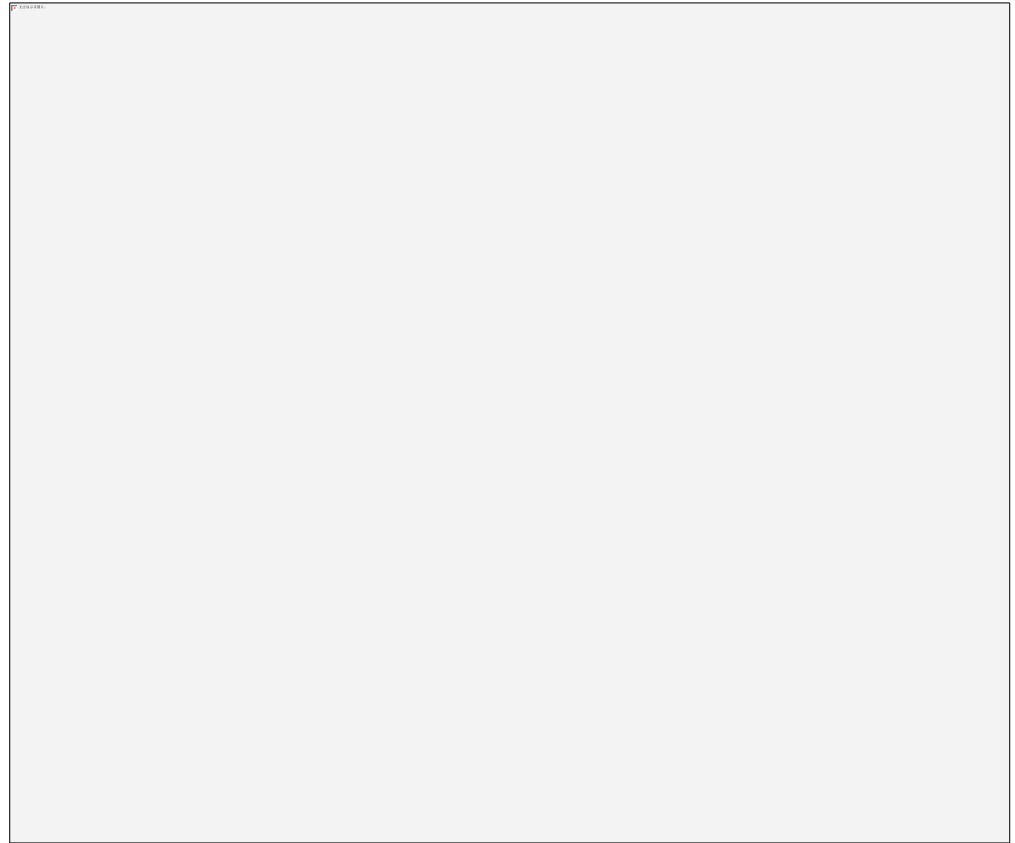
Definition of Piezoelectric Coefficients and Directions

- ✦ Orthogonal system describing the properties of a poled piezoelectric ceramic.
- ✦ Axis 3 is the poling direction.
- ✦ Because of the anisotropic nature of Piezoceramics, effects are dependent on direction.



Definition of Piezoelectric Coefficients and Directions

- ✦ To identify directions the axes, termed 1, 2, and 3, are introduced (analogous to X, Y, Z of the classical right hand orthogonal axial set).
- ✦ The axes 4, 5 and 6 identify rotations (shear).



- ✦ The direction of polarization (3 axis) is established during the poling process by a strong electrical field applied between two electrodes.
- ✦ For actuator applications the piezo properties along the poling axis are most essential (largest deflection).
- ✦ The piezoelectric coefficients described here are not independent constants.
- ✦ They vary with **temperature, pressure, electric field, form factor, mechanical and electrical boundary conditions etc.**
- ✦ The coefficients only describe material properties under small signal conditions.

Coefficients characterizing Piezoelectric materials

- ✦ **d_{ij} : Strain coefficients [m/V]:** strain developed (m/m) per electric field applied (V/m) **or** (due to the sensor / actuator properties of Piezo material).

Charge output coefficients [C/N]: charge density developed (C/m^2) per given stress (N/m^2).

- ✦ **g_{ij} : Voltage coefficients or field output coefficients [Vm/N]:** open circuit electric field developed (V/m) per applied mechanical stress (N/m^2) **or** (due to the sensor / actuator properties of Piezo material) strain developed (m/m) per applied charge density (C/m^2).

- ✦ **k_{ij} : Coupling coefficients [no Dimensions].**

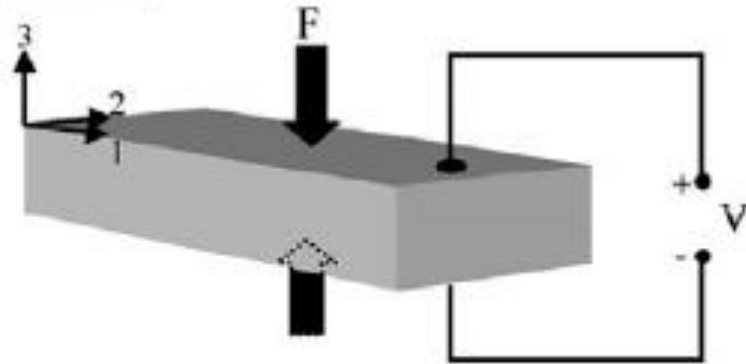
The coefficients are energy ratios describing the conversion from mechanical to electrical energy or vice versa. k^2 is the ratio of energy stored (mechanical or electrical) to energy (mechanical or electrical) applied.

- ✎ Other important parameters are the Young's modulus (*describing the elastic properties of the material*) and the dielectric constant (*describing the capacitance of the material*).
- ✎ To link electrical and mechanical quantities double subscripts (i.e. d_{ij}) are introduced.
- ✎ The first subscript gives the *direction of the excitation*,
- ✎ the second describes the *direction of the system response*.

- ✎ There are two practical coupling modes exist; the -31 mode and the -33 mode.
- ✎ In the -31 mode, a force is applied in the direction perpendicular to the poling direction, an example of which is a bending beam that is poled on its top and bottom surfaces.
- ✎ d_{31} applies if the electric field is along the polarization axis (direction 3), but the strain is in the 1 axis (orthogonal to the polarization axis).

- ✪ In the -33 mode, a force is applied in the same direction as the poling direction, such as the compression of a piezoelectric block that is poled on its top and bottom surfaces.
- ✪ d_{33} applies when the electric field is along the polarization axis (direction 3) and the strain (deflection) is along the same axis.
- ✪ Conventionally, the -31 mode has been the most commonly used coupling mode: however, the -31 mode yields a lower coupling coefficient, k , than the -33 mode.

33 Mode



31 Mode

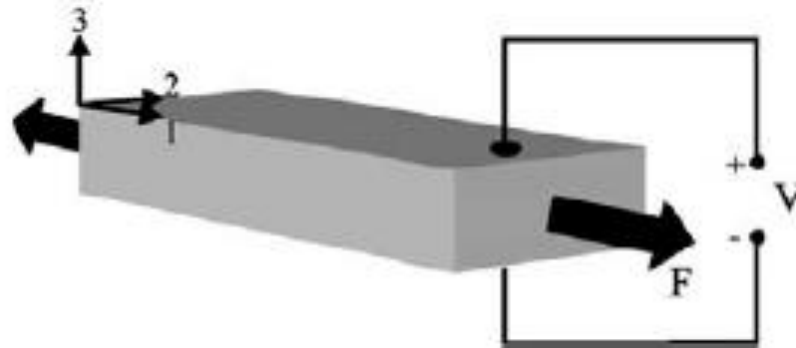
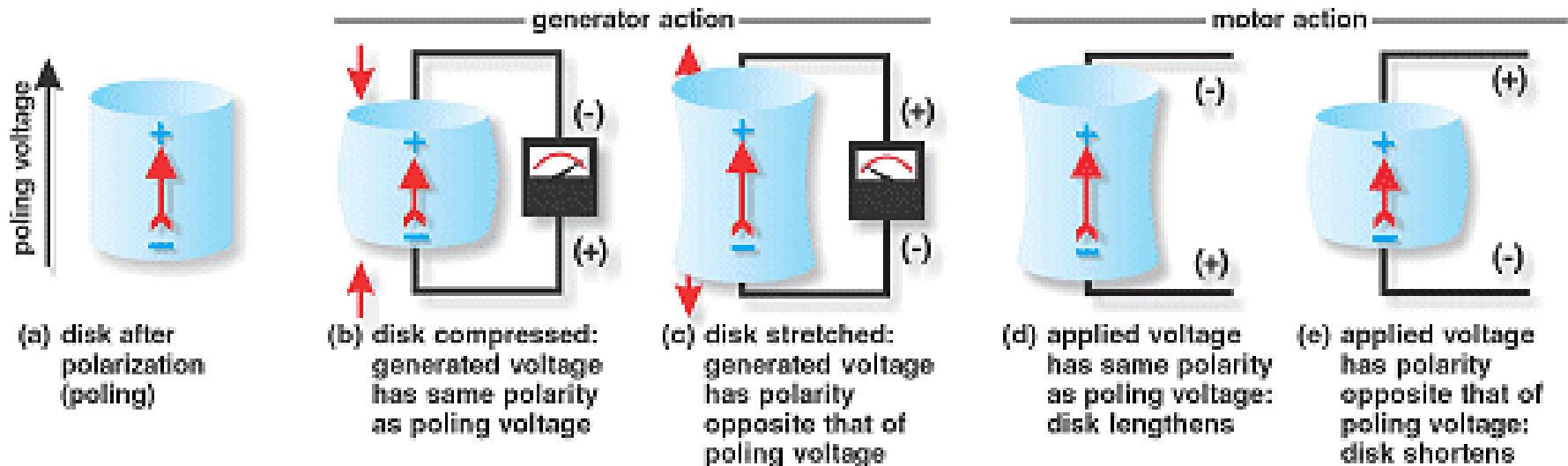


Illustration of -33 mode and -31 mode operation for piezoelectric materials. (Figure from Roundy *et al* 2003, © 2003, Elsevier.)

- ✪ It was found that in a small force, low vibration level environment, the -31 configuration cantilever proved most efficient, but in a high force environment, such as a heavy manufacturing facility or in large operating machinery, a stack configuration would be more durable and generate useful energy.
- ✪ Also found that the resonant frequency of a system operating in the -31 mode is much lower, making the system more likely to be driven at resonance in a natural environment, thus providing more power.

What can piezoelectric ceramics do?

- Mechanical compression or tension on a poled piezoelectric ceramic element changes the dipole moment, creating a voltage.
- Compression along the direction of polarization, or tension perpendicular to the direction of polarization, generates voltage of the same polarity as the poling voltage.



Generator and motor actions of a piezoelectric element

Piezoelectric ceramics- applications

- ✦ The principle is adapted to piezoelectric motors, sound or ultrasound generating devices, and many other products.
- ✦ **Generator action** is used in fuel-igniting devices, solid state batteries, and other products;
- ✦ **Motor action** is adapted to piezoelectric motors, sound or ultrasound generating devices, and many other products.

- ✪ It was found that in a small force, low vibration level environment, the -31 configuration cantilever proved most efficient, but in a high force environment, such as a heavy manufacturing facility or in large operating machinery, a stack configuration would be more durable and generate useful energy.
- ✪ Also found that the resonant frequency of a system operating in the -31 mode is much lower, making the system more likely to be driven at resonance in a natural environment, thus providing more power.

- ✪ Flexible piezoelectric materials are attractive for **power harvesting applications** because of their ability to withstand large amounts of strain.
- ✪ Larger strains provide more mechanical energy available for conversion into electrical energy.
- ✪ A second method of increasing the amount of energy harvested from a piezoelectric is to utilize a more efficient coupling mode.

<http://www.doitpoms.ac.uk/tlplib/piezoelectrics/applications.php>